Digital High-Frequency Chip — Linear Low-Dropout Voltage Regulator

Students: Arsen Arutyunov, Or Fahima

Supervisor: Tal Elazar

July 21, 2025

Abstract:

LDOs are critical components in system-on-chips because of their small layout footprint and their ability to provide a precise voltage that is independent of supply voltage and load current variations. This work presents the design, simulation, layout and verification stages of a low-dropout (LDO) voltage regulator tailored for a digitally controlled phase-locked loop (DPLL) in TSMC 28nm CMOS technology. The LDO regulates a 0.9V output from a 1.8V supply and, through two enable signals, can also deliver 0V or pass through 1.8V. It is designed to meet the demanding requirements of a noise-sensitive, timing-critical DPLL, where voltage precision is critical for minimizing jitter and ensuring accurate frequency synthesis. As a result, tight line and load regulation, along with high power supply rejection, are essential design goals. The project was carried out under a strict timeline in collaboration with peer teams developing complementary DPLL blocks, with the full system scheduled for tape-out in August 2025.

The post-layout LDO delivers up to 50 mW of load power while consuming only 80 μA of quiescent current across its analog control components, including a bandgap reference (BGR), feedback network, and two operational transconductance amplifiers (OTAs). The regulation loop is driven by a two-stage Miller-compensated OTA, which features a differential input pair and common-source output stage. This OTA achieves a DC gain of 77.6 dB, unity gain bandwidth of approximately 5.6 MHz, and a phase margin of 74°, using a nulling resistor and compensation capacitor to ensure frequency stability.

The BGR employs a current-summing topology in which PTAT and CTAT currents, with closely matched temperature coefficients, are combined in a third branch and scaled by a resistor to generate a 0.72 V reference voltage. This design achieves excellent line regulation and minimal temperature drift without relying on second-order temperature compensation techniques or advanced current mirrors.

The LDO exhibits robust regulation characteristics with post-layout simulation results showing a line regulation of 0.0009 V/V, load regulation of 0.033 mV/mA, and temperature coefficient (TC) of 6.4 ppm/°C, verified across multiple corners. Power supply rejection ratio (PSRR) at 1 kHz is approximately −61 dB. The transient response was validated by applying a 10 ps-wide disturbance on the supply voltage and load current, with the output stabilizing within $1 - 10\mu S$ at worst case, which demonstrates the regulator's ability to suppress fast digital switching noise. The full LDO feedback loop maintains a phase margin of 58°, which ensures overall stability across operating conditions.

The layout stage was implemented following industry best practices, including usage of short and straight wires for current branches, common-centroid placement for OTA differential pairs and BJTs in the BGR, symmetry-based matching of devices and wires, and metal stacking to minimize IR drop in the LDO. All in all, the project successfully demonstrates the integration of an LDO with excellent TC and line and load regulation suitable for the DPLL architecture.

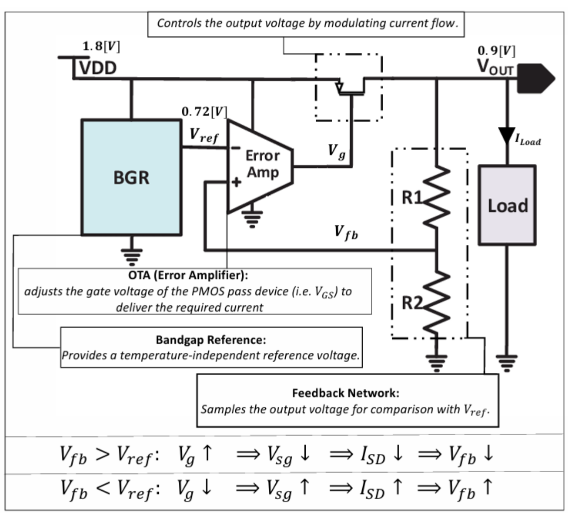

Figure 1. Block Diagram of the Designed LDO with Integrated OTA and BGR

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1 Introduction

1.1 Goals of the Project

The main goal of this project was to fully design, simulate, implement layout and validate from scratch a linear low-dropout (LDO) voltage regulator in TSMC 28nm technology. The LDO should supply stable and accurate power to three critical digital blocks of a phase-locked loop (DPLL) system, including a DCO and TDC. These blocks, developed by other collaborating teams, require a clean and tightly regulated 0.9V supply derived from a 1.8V input. Since the DPLL is highly sensitive to voltage fluctuations, especially in its digitally controlled oscillator, it is essential to have tight line regulation, low temperature drift, and excellent noise immunity to minimize jitter and maintain signal integrity. The LDO is designed to deliver a constant power of 50mW, corresponding to a load current of approximately 55.5mA.

Unlike power supplies intended for highly dynamic loads, the DPLL presents a relatively steady DC load. As such, the LDO design emphasizes voltage accuracy, thermal robustness, and fast response to supply disturbances, rather than prioritizing load regulation. Transient response remains relevant due to fast digital switching in nearby blocks that may introduce noise on the power rails. The LDO must suppress these disturbances to maintain regulation integrity.

To meet the stringent performance demands of the system, the following design goals were set at the beginning:

- **Output voltage of BGR:** 0.72V from a 1.8V supply.
- **Output voltage of LDO:** 0.9V from a 1.8V supply.
- **Load current drawn from LDO:** 55.5mA (50mW).
- **Line regulation:** ≤ 0.01V/V (-40dB).
- **Load regulation:** ≤ 0.1mV/mA from 0–60mA
- **Temperature coefficient (TC):** ≤ 60ppm/°C across -40°C to 125°C.
- **PSRR:** ≤ -40dB at 1kHz.
- **Quiescent current:** ≤ 1mA (shared across both OTAs, the BGR, and feedback network).
- **OTA performance:** ≥ 70dB DC gain and ≥5MHz GBW for feedback response accuracy and speed.

In addition to circuit design, careful attention was given to the layout to ensure the LDO performs well after fabrication. Common-centroid placement was used for matched transistors in both the OTA and the BGR to reduce mismatch caused by systematic gradients and to improve robustness against process, voltage, and temperature (PVT) variations. In the OTA, dummy devices were added to improve symmetry and matching. Symmetrical metal routing was applied at sensitive nodes to balance parasitics and improve performance. For the LDO current path, metal stacking and symmetrical layout were used to reduce IR drop and ensure stable output voltage under varying conditions. The layout was planned not only to match post-layout simulation goals, but also to maintain performance after fabrication.

1.2 Motivation

Digitally controlled Phase-Locked Loops (DPLLs) are critical components in modern mixed-signal systems, often used for clock generation, data recovery, and frequency synthesis in communication and digital processing chips. These systems are highly sensitive to fluctuations in supply voltage, which can introduce jitter and degrade timing accuracy. As such, the power supply for a DPLL must offer exceptional stability, low noise, and minimal variation across temperature, line and load changes to ensure high signal integrity and frequency precision.

A Linear Low-Dropout (LDO) voltage regulator is particularly well-suited for this application due to its key advantages in noise performance, simplicity, and integration. Unlike switching regulators such as DC-DC buck converters or charge-pump-based solutions, LDOs do not rely on inductive or capacitive energy storage elements that typically require large passive components. This makes LDOs significantly more compact in layout—especially important in advanced CMOS nodes like TSMC 28nm, where silicon area is costly

Furthermore, the aforementioned switching regulators inherently generate high-frequency noise due to their switching operation, which can couple into sensitive analog nodes and degrade the performance of noise-sensitive circuits such as the digitally-controlled oscillator (DCO) and phase detector in the DPLL. In contrast, LDOs operate in a continuous, linear fashion, offering low output ripple and minimal high-frequency noise, which is essential for clean clock generation and low phase noise in the DPLL. However, because LDOs operate using linear (resistive) regulation, they have low power efficiency when the input voltage V_{DD} is much higher than the output voltage V_{out}, and the excess power is dissipated as heat. For more see [6].

Charge-pump regulators, while sometimes used in DPLL architectures themselves, are unsuitable as power supplies due to their reliance on large capacitors, limited drive capability, and sensitivity to leakage and switching mismatches. They also struggle with load regulation over wide ranges, making them incompatible with the constant power demand of multiple DPLL blocks.

In our system, the LDO is responsible for powering three internal digital and mixed-signal blocks of the DPLL. These blocks have a relatively constant and well-defined current draw of approximately 55.5 mA (50 mW at 0.9 V), allowing the LDO to be optimized for a narrow operating point. High power supply rejection ratio (PSRR) and excellent line regulation are prioritized to prevent external supply ripple from propagating into the DPLL's timing-critical circuitry.

Overall, since the LDO requires minimal external components and no inductors, they are ideal for on-chip integration with a DPLL because they offer the optimal balance of low noise, compact layout, fast transient response, and design simplicity.

1.3 The Approach to Solving the Problem

The project began with a detailed study of low-dropout (LDO) regulators, bandgap references (BGRs), and operational amplifier design. After reviewing the theory, we selected a BGR topology that initially used voltage summation of PTAT and CTAT voltages to generate a reference of around 1.2V with a small temperature coefficient. For the LDO's error amplifier, we chose a two-stage Miller-compensated OTA to meet our gain and stability requirements.

We first implemented the BGR using a current mirror configuration since it is simpler, but the TC and PSRR was poor due to voltage mismatch between branches. To overcome this, we replaced the current mirror with an ideal op-amp model provided in the simulation environment. This improved the performance significantly, allowing the BGR to reach a TC-stable output around 1.2V.

Next, we designed the two-stage Miller OTA, tuning its gain and compensation components to achieve a GBW of 5–10MHz and a phase margin of over 60°. After verifying its performance independently, we integrated it with the BGR to confirm compatibility. Once the system worked, we modified the BGR topology to generate a final reference voltage of 0.72V by summing PTAT and CTAT currents instead of voltages, while maintaining low temperature drift.

Before moving to layout, we confirmed in simulation that the complete LDO design functioned as intended, with proper regulation and stability. Then, we moved on to layout implementation. We started with the OTA, and went through several iterations to improve device matching and symmetry. Dummy devices and common-centroid placement were used to enhance performance. Once it passed LVS and DRC, and post-layout simulations confirmed proper function, we proceeded to the BGR.

The BGR layout also went through multiple refinements. Careful placement of BJTs in common-centroid configuration and attention to routing symmetry were used to minimize mismatch.

Finally, we constructed the full LDO layout, and integrated the OTA, BGR, and pass device. Emphasis was placed on minimizing parasitic resistance in the current path and achieving good symmetry to reduce IR drop. After additional design tweaks, the LDO met its key targets including line and load regulation, PSRR, and stability. Each layout stage passed layout-versus-schematic (LVS), design rule checks (DRC), and antenna effect checks.

The entire project workflow ensured that each block was verified before integration through rigorous simulation cycles, and the final layout checks validated the design's manufacturability and functional correctness.

1.4 Comparison Against Existing Implementations

Existing LDO regulator implementations found in the literature or commercially are often optimized for general-purpose use and are not tailored to meet the specific requirements of powering a DPLL system-on-chip. These designs typically have large layout footprints due to off-chip or non-area-constrained applications, making them unsuitable for integration in compact SoC environments. Moreover, their output voltage (V_{OUT}), supply voltage range (V_{DD}), and supported load current (I_{LOAD}) are frequently much higher and mismatched with the demands of this DPLL system.

Another significant limitation lies in the technology compatibility. Many of these regulators are implemented in older or incompatible process nodes, making direct integration with a 28 nm TSMC-based DPLL chip impractical or inefficient. Additionally, many LDO implementations often exhibit suboptimal line regulation and degraded performance over wide temperature ranges, particularly from −40 °C to 125 °C, which can be critical in precision timing applications.

In this work, a custom LDO designed in 28nm is presented to address the aforementioned shortcomings, with an emphasis on minimizing layout area and ensuring robust operation across process, voltage, and temperature (PVT) variations. The design also features specially tailored enable signals, aligned with the system-level requirements of the DPLL, for reliable control purposes. A detailed quantitative comparison between the proposed LDO and existing implementations, including key performance metrics such as PSRR, load/line regulation, and temperature stability, will be presented in the *Results and Analysis* section.

2 Theoretical Background

2.1 Low-Dropout Voltage Regulator

LDO Design Principles, Tradeoffs, and Stability A typical analog LDO (Fig. 1) comprises a bandgap reference, an error amplifier (EA), a resistive feedback network, and a PMOS pass transistor. The bandgap generates a temperature- and supply-independent reference voltage. The EA compares this reference to a scaled output voltage and adjusts the PMOS gate voltage to regulate the output current and voltage.

Assuming an ideal EA with infinite input impedance and a virtual short, the output voltage is:

$$V_{\text{out}} = V_{\text{ref}} \left(1 + \frac{R_{\text{fb1}}}{R_{\text{fb2}}} \right). \tag{1}$$

An analog LDO with an OTA-based EA is preferred over digital LDOs for better PSRR, lower noise, and design simplicity, although digital LDOs offer better process portability and lower power. The OTA gain must be sufficiently high (typically > 70 dB) to enforce a virtual short and ensure precise regulation.

The pass transistor usually operates in saturation mode for higher loop gain and PSRR, despite triode mode enabling lower dropout voltage ($V_{\text{DO}} = V_{DD} - V_{out}$). PMOS devices are favored since NMOS requires charge pumps for gate drive above the supply voltage, but PMOS adds stability challenges due to gate capacitance.

Dynamic performance metrics—PSRR, output noise, and transient response—are as critical as static line/load regulation. Stability is complicated by load-dependent output poles and dominant gate poles, necessitating output capacitors (often with series damping resistors) to maintain phase margin and improve PSRR at the cost of area and bandwidth.

Key Performance Metrics

- **Temperature Coefficient (TC)** (ppm/°C): measures output voltage variation over temperature,

$$\text{TC} = \frac{V_{\max} - V_{\min}}{V_{\text{avg}} \cdot \Delta T} \times 10^6.$$

- **Line Regulation** (V/V): output voltage sensitivity to supply voltage changes (at constant load),

$$\text{Line Regulation} = \frac{\Delta V_{\text{out}}}{\Delta V_{DD}}.$$

- **Load Regulation** (mV/mA): output voltage sensitivity to load current variations (at constant supply),

$$\text{Load Regulation} = \frac{\Delta V_{\text{out}}}{\Delta I_{\text{load}}}.$$

- **Power Supply Rejection Ratio (PSRR)** (dB): ratio of output voltage ripple to supply ripple,

$$\text{PSRR} = 20\log_{10}\left(\frac{\Delta v_{\text{out}}}{\Delta v_{DD}}\right).$$

  Typical commercial bandgap LDOs (Table 1) exhibit TC between 10–100 ppm/°C, line regulation on the order of 10^{-4} V/V, and load regulation around 0.01–0.1 mV/mA.
- **Quiescent Current I_Q:**

$$I_Q = I_{\text{total}} - I_{\text{load}},$$

  representing current consumed by the LDO circuitry, minimized by large feedback resistors and efficient OTA/BGR design.
- **Dropout Voltage V_{DO}:**

$$V_{DO} = V_{DD} - V_{out} = I_{\text{load}} \cdot R_{DS(on)} \approx V_{DS} \quad \text{(pass transistor).}$$

  Lowering V_{DO} by increasing transistor size trades off with increased gate parasitic capacitance, impacting stability.
- **Power Efficiency η:**

$$\eta = \frac{V_{\text{out}} I_{\text{load}}}{V_{DD}(I_{\text{load}} + I_Q)} = \frac{(V_{DD} - V_{DO}) I_{\text{load}}}{V_{DD}(I_{\text{load}} + I_Q)}.$$

- **Input/Output Range:** defines supply voltage and load current limits for stable regulation.
- **Transient Response:** depends on OTA loop gain and output filter, determining settling time and overshoot.
- **Loop Gain and Stability Metrics:**
  - *Phase Margin (PM)*: phase difference from −180° at unity gain; 45°–90° indicates stability.
  - *Gain Margin (GM)*: gain difference in dB at −180° phase crossing; > 10–20 dB preferred.
  - *Gain-Bandwidth Product (GBW)*:

$$GBW = |A_{DC,OL}| \cdot f_{-3dB}.$$

- **Output Noise:** dominated by OTA flicker noise at low frequencies,

$$\overline{V^2_{\text{noise,out}}} = \left(1 + \frac{R_1}{R_2}\right)^2 \overline{V^2_{\text{noise,OTA}}}.$$

- **Figure of Merit (FOM):**

$$FOM = \frac{C_{\text{out}}\, \Delta V_{\text{out}}\, I_Q}{I^2_{\text{load,max}}},$$

  measuring transient regulation efficiency; lower values are better.

Parameter	NCP4681	ADP121	LD39015
TC (ppm/°C)	100	10	30
Load Regulation (mV/mA)	0.268	0.01	0.02
Line Regulation (V/V)	0.0002–0.001	0.0003	0.0001
PSRR (dB)	@1kHz - 25	@10kHz - 66	@1kHz - 65

Table 1
Typical Commercial LDO Parameters [28, 29, 30]

Line Regulation Analysis Using the LDO model in Fig. 2, the OTA controls the pass transistor gate-source voltage:

$$V_{SG} = V_{DD} - A\beta(V_{OUT} - V_{REF}),$$

where A is the EA gain, and $\beta = \frac{R_{fb2}}{R_{fb1}+R_{fb2}}$ the feedback factor.

Small-signal perturbations satisfy:

$$\Delta V_{SG} = \Delta V_{DD} - A\beta\Delta V_{OUT}.$$

Expressing output voltage change due to supply variation:

$$(\Delta V_{DD} - \beta A \Delta V_{OUT}) g_m (R_{Load} \parallel r_{DS} \parallel (R_{fb1} + R_{fb2})) = \Delta V_{OUT}.$$

Solving for line regulation:

$$\frac{\Delta V_{OUT}}{\Delta V_{DD}} = \frac{g_m (R_{Load} \parallel r_{DS} \parallel (R_{fb1} + R_{fb2}))}{1 + \beta A g_m (R_{Load} \parallel r_{DS} \parallel (R_{fb1} + R_{fb2}))} \approx \frac{1}{\beta A}. \quad (2)$$

This shows that line regulation and PSRR improve with higher OTA gain and feedback factor.

Load Regulation Analysis Load current variations produce:

$$\Delta I_{Load} = \frac{\Delta V_{OUT}}{r_{DS} \parallel (R_{fb1} + R_{fb2}) \parallel R_{Load}} + \Delta V_{OUT} \cdot \beta A g_m,$$

where the first term is passive current through output impedance and feedback, and the second term is active correction.

Solving for load regulation:

$$\frac{\Delta V_{OUT}}{\Delta I_{Load}} = \frac{r_{DS} \parallel (R_{fb1} + R_{fb2}) \parallel R_{Load}}{1 + \beta A g_m (r_{DS} \parallel (R_{fb1} + R_{fb2}) \parallel R_{Load})} \approx \frac{1}{\beta A g_m}. \quad (3)$$

Since g_m is typically limited (e.g., < 1 S), load regulation is generally worse than line regulation. For $\beta = 0.8$, OTA gain 62 dB, and $g_m = 100$ mS, load regulation ≈ 0.01 V/A. Improving beyond this requires higher OTA gain (e.g., 80 dB).

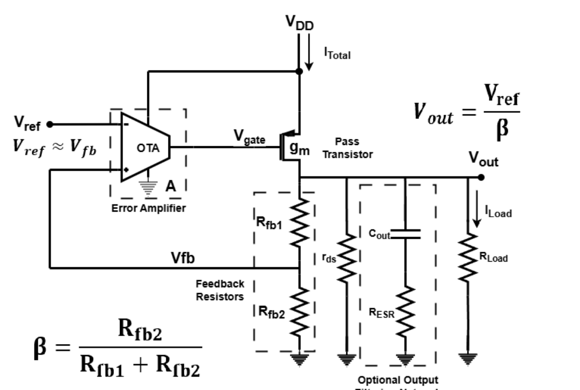

Figure 2. LDO Circuit for Line/Load Regulation and Stability Analysis

Loop Gain and Stability Analysis

The feedback loop gain of an LDO with a two-stage OTA is shaped by three dominant poles at high-impedance nodes:

$$\omega_{p1} = \frac{1}{R_{\text{out}} C_{\text{out}}} = \frac{1}{(r_{\text{ds}} \parallel R_{\text{Load}} \parallel (R_{fb1} + R_{fb2}))C_{\text{out}}}$$
$$\omega_{p2} = \frac{1}{R_{\text{out,OTA}} C_{G,\text{PMOS}}}$$

where C_{out} is the output decoupling capacitor (on- or off-chip), r_{ds} the transistor output resistance, R_{Load} the load resistance, and $R_{\text{out,OTA}}$, $C_{G,\text{PMOS}}$ are the OTA output resistance and PMOS gate capacitance respectively. The latter increases with pass transistor width to support higher load currents.

The dominant pole ω_{p1} depends on the load current I_{Load} as:

$$\omega_{p1} \approx \frac{1}{r_{\text{ds}} C_{\text{out}}} = \frac{I_{\text{Load}}}{V_A C_{\text{out}}}$$

where V_A is the Early voltage, and C_{out} is assumed constant. As load current increases, the output pole shifts to higher frequencies, potentially crossing other poles and destabilizing the loop. To mitigate this, a series damping resistor is often added with the output capacitor to introduce a left-half-plane (LHP) zero, improving phase margin.

The loop gain LG is expressed as:

$$LG = A \cdot \beta \cdot g_m \cdot R_{\text{out}} = A \cdot \beta \cdot g_m \cdot \left(r_{ds} \parallel R_{\text{Load}} \parallel (R_{fb1} + R_{fb2}) \right)$$

with A the error amplifier gain, $\beta = \frac{R_{fb2}}{R_{fb1}+R_{fb2}}$ the feedback factor, and g_m the pass transistor transconductance.

The open-loop transfer function $T(s)$ is:

$$T(s) = \frac{A\beta g_m (r_{ds} \parallel R_{\text{Load}} \parallel (R_{fb1} + R_{fb2})) \left(1 + \frac{s}{\omega_z}\right)}{\left(1 + \frac{s}{\omega_{p1}}\right)\left(1 + \frac{s}{\omega_{p2}}\right)\left(1 + \frac{s}{\omega_{p3}}\right)}$$

Equation 1: Open-loop Transfer Function of the LDO

where ω_z is the OTA's internal compensation zero frequency, and V_+, V_{fb} denote the OTA input and feedback node voltages respectively.

Design Implications

In the DPLL chip context, the output capacitance is dominated by small decoupling capacitors, totaling no more than approximately 300 fF. This leads to a very high output pole frequency (1 GHz), which does not compromise stability. Therefore, the LDO stability primarily depends on the OTA design, and no dedicated output capacitor or series resistor is included in the LDO layout.

Stability must be verified over the expected range of output capacitance, and simulations should analyze trade-offs between capacitor sizes. Inclusion of a series damping resistor and additional output capacitance is deferred to the top-level system design, based on full integration performance results.

2.2 OTA as an Error Amplifier

Role in LDO and Bandgap Reference (BGR) The Operational Transconductance Amplifier (OTA) is the core high-gain analog block enabling closed-loop regulation in both the LDO and the bandgap reference (BGR). In the LDO, it compares a scaled output voltage to a reference and adjusts the pass transistor gate to stabilize the output under varying load and supply conditions. In the BGR, it balances voltages across two branches to generate a Proportional To Absolute Temperature (PTAT) current, combined with a Complementary To Absolute Temperature (CTAT) component, producing a temperature-stable reference voltage.

Why an OTA? Unlike conventional voltage amplifiers, the OTA outputs a current proportional to its differential input voltage. While designed as a current-output device with high output impedance driving capacitive loads, in LDO and BGR applications the OTA effectively operates as a voltage amplifier by charging/discharging capacitive nodes. Thus, the overall loop gain and regulation rely heavily on the OTA's transconductance gain.

Architecture and Gain Requirements The OTA employs a two-stage topology (Fig. 3):

- A differential input stage providing initial gain,
- A common-source output stage boosting gain and driving capacitive loads.

This design achieves high open-loop gain (typically > 60 dB), critical for:

- Improved line and load regulation in the LDO,
- Enhanced node voltage matching, temperature stability, and PSRR in the BGR.

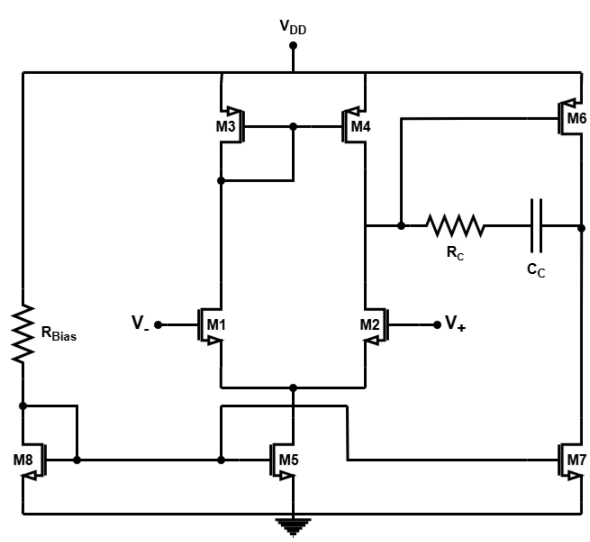

Figure 3. Two-Stage Miller OTA Topology

Stability and Pole Splitting Two-stage OTAs inherently have two close poles, risking low phase margin and oscillations. The first stage's output pole is

$$\omega_{p1} = \frac{1}{(r_{ds2} \parallel r_{ds4})C_{\text{par}}},$$

and the second stage's pole is

$$\omega_{p2} = \frac{1}{(r_{ds6} \parallel r_{ds7})C_{\text{Load}}}.$$

Similar transistor sizing and bias currents cause these poles to cluster, reducing stability.

Pole splitting is achieved by introducing a compensation capacitor C_c between the two stages, shifting the dominant pole p_1 to lower frequencies and pushing p_2 higher, often aided by increasing the second stage transconductance g_{m6} (via transistor sizing or bias current). This ensures sufficient gain roll-off before excessive phase shift accumulates.

Miller Effect and Trade-offs Miller compensation leverages the negative gain of the second stage to amplify C_c, yielding an effective input capacitance:

$$C_{\text{in,eff}} = C_c(1 + |A_v|),$$

where A_v is the voltage gain magnitude. This shifts p_1 to lower frequencies, enhancing phase margin but reducing bandwidth, as shown in Fig. 4.

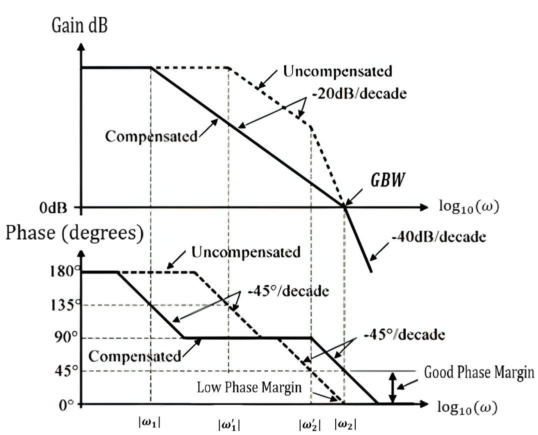

Figure 4. Bode Plot: Compensated vs. Non-compensated OTA

While C_c minimally affects the non-dominant pole (Fig. 5), it introduces a right-half-plane (RHP) zero due to a feedback path at high frequencies, reducing phase margin. The zero frequency decreases with increasing C_c, potentially degrading stability near unity gain.

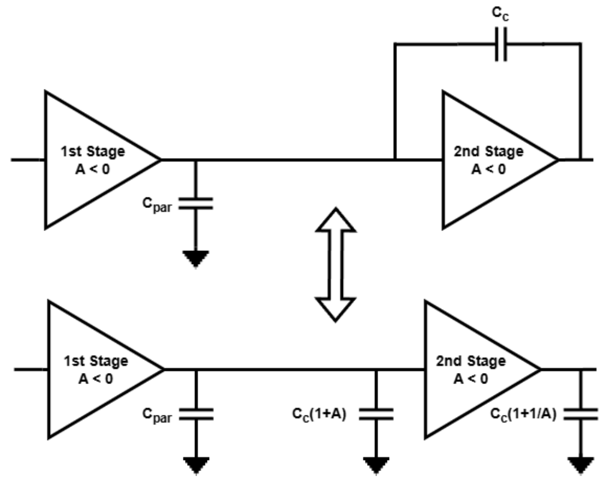

Figure 5. Miller Capacitor C_c Equivalent Model

Adding a series resistor R_c with C_c shifts this zero to the left half-plane (LHP) or to higher frequencies in the RHP, improving phase margin with minimal impact on poles:

$$z = \frac{1}{C_c\left(\frac{1}{g_{m2}} - R_c\right)} \quad \text{(LHP zero if } R_c > \frac{1}{g_{m2}}\text{).}$$

Small-Signal Transfer Function Using the small-signal model in Fig. 6, neglecting R_c, the OTA voltage gain is derived as:

$$\frac{v_{\text{out}}}{v_{\text{in}}} = \frac{g_{m1}R_1g_{m2}R_2\left(1 - \frac{sC_c}{g_{m2}}\right)}{s^2R_1R_2(C_{\text{par}}C_c + C_{\text{par}}C_L + C_cC_L) + s\left[R_2(C_c + C_L) + R_1(C_c + C_{\text{par}}) + g_{m2}R\right.}$$

It exhibits two poles and one zero, where the dominant pole p_1 approximates to:

$$p_1 \approx \frac{1}{g_{m2}R_2C_cR_1},$$

and the gain-bandwidth product (GBW) is:

$$GBW = g_{m1}/C_c.$$

The zero is at:

$$z = \frac{g_{m2}}{C_c},$$

confirming the zero frequency lowers with larger C_c, worsening stability without compensation.

Summary The OTA design balances high gain, phase margin, and stability critical for LDO and BGR performance. Miller compensation with a series resistor stabilizes the amplifier with minimal area and power overhead, enabling robust integration within the DPLL system.

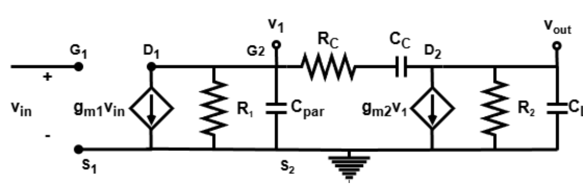

Figure 6. Small-Signal Model of the Two-Stage Miller OTA

2.3 Bandgap Reference Circuit

Motivation Low-dropout regulators (LDOs) require precise, temperature-stable voltage references to maintain consistent output despite variations in temperature, supply voltage (V_{DD}), and process parameters. Traditional references (passive dividers, MOS threshold, diode-based) suffer from poor temperature stability or process sensitivity.

The Bandgap Reference (BGR) is widely favored due to its excellent temperature compensation and insensitivity to supply and process variations, making it essential for precision analog/mixed-signal circuits such as LDOs, ADCs, and DACs.

Operation Principle BGRs typically combine two voltages with opposing temperature coefficients over a wide range (e.g., $-40°C$ to $125°C$):

- A **CTAT** (complementary-to-absolute-temperature) voltage, usually the BJT base-emitter voltage V_{EB}, which decreases with temperature ($-1.5\,\text{mV}/°\text{C}$).
- A **PTAT** (proportional-to-absolute-temperature) voltage, derived from the thermal voltage $V_T = \frac{kT}{q}$, increasing approximately $86\,\mu\text{V}/°\text{C}$.

By appropriately scaling and summing these voltages, the BGR achieves a near-zero temperature coefficient reference voltage:

$$V_{\text{ref}} = \alpha \cdot V_T + \beta \cdot V_{EB} \quad \text{with} \quad \frac{dV_{\text{ref}}}{dT} \approx 0.$$

The base-emitter voltage of a forward-active BJT is:

$$V_{EB} = V_T \ln\left(\frac{I_C}{I_S}\right),$$

where I_C is the collector current and I_S is the saturation current, strongly temperature-dependent due to intrinsic carrier concentration and diffusion effects.

Assuming constant I_C, the temperature coefficient of V_{EB} approximates:

$$\frac{dV_{EB}}{dT} \approx \frac{V_{EB} - 2.5V_T - \frac{E_g}{q}}{T},$$

where E_g is silicon's bandgap energy. More accurate analysis considers the PTAT nature of I_C, slightly adjusting the coefficient (e.g., from -1.6 to $-1.5\,\text{mV}/°C$ at room temperature).

Circuit Implementation and Scaling Key implementation details include:

1. **Current Matching:** Equal collector currents in two BJTs are ensured via a current mirror (transistors M_1, M_2, M_3) with matched transistor geometries and biasing to reduce Early effect and mismatch.
2. **Voltage Matching:** An OTA with high gain enforces equal voltages ($V_A = V_B$) at the BJT bases, enabling accurate voltage difference sensing.
3. **Emitter Area Ratio:** Two BJTs with emitter areas differing by a factor N generate a PTAT voltage:

$$V_{R_1} = V_{EB1} - V_{EB2} = V_T \ln(N),$$

producing a PTAT current:

$$I_C = \frac{V_T \ln(N)}{R_1}.$$

Selecting a resistor R_1 with low temperature coefficient (e.g., polysilicon) minimizes error. Typically, $N = 8$ balances noise immunity and layout area.

Reference Voltage Formation Adding a third BJT and resistor R_2 produces the final temperature-compensated voltage:

$$V_{\text{ref}} = \frac{R_2}{R_1} \ln(N) V_T + V_{EB},$$

with the coefficient $\alpha = \frac{R_2}{R_1} \ln(N)$ chosen to satisfy:

$$\alpha \frac{dV_T}{dT} + \frac{dV_{EB}}{dT} = 0.$$

This yields $V_{\text{ref}} \approx 1.16\,\text{V}$, near silicon's bandgap voltage, with near-zero temperature dependence (see Figure 7).

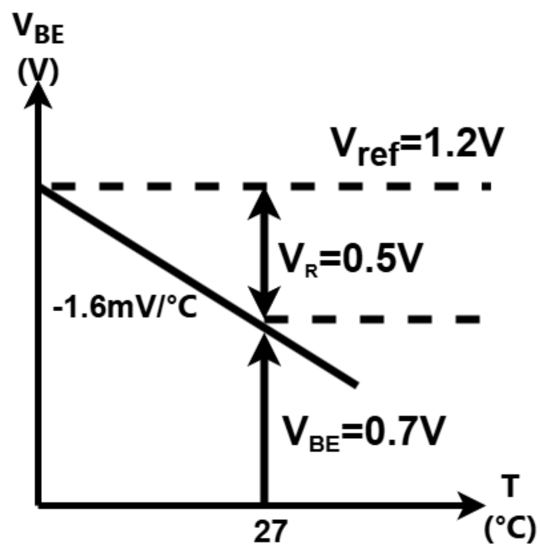

Figure 7. Temperature Compensation in a Bandgap Reference: Combining CTAT and PTAT Components to achieve a stable ~1.2 V output.

Feedback Loop Stability The OTA operates in DC with high open-loop gain, driving the difference between nodes V_A and V_B to zero, enabling stable PTAT current generation. Proper input polarity ensures negative feedback: an increase in $V_A - V_B$ leads to OTA output changes that reduce this difference via current mirror transistor operation.

Stability is maintained by ensuring the negative feedback gain dominates the positive feedback, typically by choosing $R_1 > 10\,\text{k}\Omega$ and using matched transistor transconductances g_m:

$$A_{\text{loop}} \approx -A_{\text{OTA}} \cdot g_m \cdot R_1.$$

Current-Mode Bandgap Reference To enable output voltage tuning beyond the fixed $\sim$ 1.2 V limit of voltage-mode BGRs, current-mode BGRs generate a temperature-independent current by summing PTAT and CTAT currents:

$$I_{\text{PTAT}} = \frac{kT}{qR_1}\ln(N), \quad I_{\text{CTAT}} = \frac{V_{EB}}{R_2}.$$

The output voltage is:

$$V_{\text{ref}} = R_3\,(I_{\text{PTAT}} + I_{\text{CTAT}}) = R_3\left(\frac{\ln(N)}{R_1}V_T + \frac{V_{EB}}{R_2}\right).$$

By selecting resistor ratios R_1, R_2, and R_3, the output voltage can be scaled and tuned while maintaining temperature compensation, allowing V_{ref} values different from the silicon bandgap voltage (Figure 8).

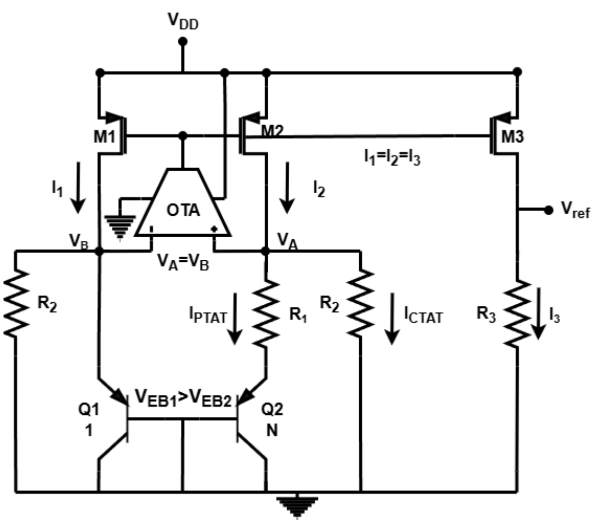

Figure 8. Current-Mode Bandgap Reference Circuit Topology

Precise temperature coefficient cancellation is refined via simulation-based iterative resistor sizing, as exact resistor TC and process effects complicate closed-form solutions.

2.4 *Transistor Operation Considerations for the LDO*

In analog blocks such as LDOs, bandgap references, and OTAs, MOSFETs predominantly operate in saturation, providing high intrinsic gain, current-source behavior largely independent of V_{DS}, and predictable linearity critical for feedback stability. While channel-length modulation causes some V_{DS}-dependence, saturation remains the preferred region for robust analog design.

Saturation can be subdivided into *weak*, *moderate*, and *strong* inversion based on the gate-source voltage V_{GS} relative to threshold V_{TH}. For simplicity, moderate inversion is often grouped with either regime due to its transitional nature. Strong inversion corresponds to $V_{GS} > V_{TH}$, weak inversion to $V_{GS} < V_{TH}$ (non-cutoff).

In **strong inversion**, the channel is pinched off near the drain, yielding drain current:

$$I_{DS} = \frac{\mu C_{ox}}{2}\frac{W}{L}(V_{GS} - V_{TH})^2(1 + \lambda V_{DS}), \qquad (??)$$

where μ is mobility, C_{ox} oxide capacitance per unit area, W and L device dimensions, and λ models channel-length modulation (Early effect).

In **weak inversion**, diffusion dominates, resulting in an exponential current dependence:

$$I_{DS} = I_0\frac{W}{L}e^{\frac{V_{GS}-V_{TH}}{nV_T}}\left(1 - e^{-\frac{V_{DS}}{V_T}}\right) \approx I_0\frac{W}{L}e^{\frac{V_{GS}-V_{TH}}{nV_T}} \quad (V_{DS} > 3V_T), \qquad (??)$$

where I_0 is process-dependent, n the subthreshold slope factor, and V_T the thermal voltage. For $V_{DS} > 3V_T$, current saturates and is effectively V_{DS}-independent.

The transconductance differs markedly between regions:

$$g_{m,\text{SI}} = \frac{\partial I_{DS}}{\partial V_{GS}} = \mu C_{ox}\frac{W}{L}(V_{GS} - V_{TH}) = \frac{2I_{DS}}{V_{GS} - V_{TH}},$$

$$g_{m,\text{WI}} = \frac{\partial I_{DS}}{\partial V_{GS}} = \frac{I_{DS}}{nV_T}.$$

Weak inversion offers inherently higher transconductance efficiency $\frac{g_m}{I_{DS}}$, independent of device geometry, and very high output resistance $r_o = \frac{V_A}{I_{DS}}$, yielding large intrinsic gain $A_v = g_m r_o$ at low bias currents. This makes weak inversion attractive for low-power analog design such as the LDO in a DPLL chip.

However, the trade-off is bandwidth: the high output resistance limits bandwidth (typically below 1 kHz), constraining speed. Additionally, weak inversion devices are more sensitive to process, voltage, and temperature variations, potentially affecting reliability.

A practical compromise is operation near the boundary $V_{GS} \approx V_{TH}$, combining high gain with improved robustness.

Despite increased flicker and thermal noise in weak inversion, the low bandwidth and filtering in LDOs suppress noise effectively, ensuring clean supply voltages for sensitive DPLL subblocks where loop bandwidths are typically low ($\sim$ 1 kHz).

3 Implementation

3.1 *Project Implementation Steps*

The LDO implementation followed a structured process:

1. **Initial BGR Topology Selection:** Selected a Bandgap Reference (BGR) topology generating $\sim$ 1.2 V. Verified basic operation with an ideal OTA model.
2. **Determining OTA Gain Requirements:** Simulations showed a minimum OTA gain of 60 dB ensures acceptable performance; gains above 80 dB offered diminishing returns.

3. **OTA Topology and Design:** Adopted a two-stage Miller-compensated OTA targeting $\geq$ 60 dB gain. Compensation capacitor was sized considering the initial pass transistor gate capacitance.
4. **Integration into BGR:** Replaced ideal OTA with the designed OTA; full loop stability and expected operation were confirmed.
5. **Scaling for Low V_{REF}:** To achieve $V_{\text{REF}} < 0.9\,\text{V}$, switched to a current-mode BGR with added current-scaling branch and adjusted resistors.
6. **LDO Implementation and PMOS Sizing:** Implemented LDO using the verified BGR and OTA. PMOS width increased to support 55.5 mA load, requiring minor compensation tuning to maintain phase margin.
7. **Output Capacitance and Stability Testing:** Verified stable performance across a range of output capacitances and series resistors under varying load conditions.
8. **Layout Process:**
   - **OTA Layout:** Focused on symmetry, common-centroid placement, and current mirror matching to preserve DC gain and minimize offset. Multiple iterations ensured design rule compliance and layout advisor approval.
   - **BGR Layout:** Simpler due to DC operation; common-centroid layout of BJTs minimized PVT variations.
   - **LDO Layout:** Minimized IR drop by optimizing high-current routing and compacting OTA layout for symmetry and ease of integration. Final layout approved after IR drop reduction.
9. **Enable Signal Integration:** Added dual enable signals to switch output between 0 V (disabled), 0.9 V (regulated), and 1.8 V (bypass), with minor schematic and layout modifications.
10. **Final Validation:** Verified reliable operation across process, voltage, and temperature (PVT) corners.

3.2 Bandgap Reference (BGR)

Current Mirror-Based Design Attempts

An initial BGR design (Fig. 9) used a current mirror to equalize voltages at nodes A and B. Despite using large channel lengths, variations in V_{DS} caused mismatches and poor PSRR (−17 dB) due to insufficient voltage matching.

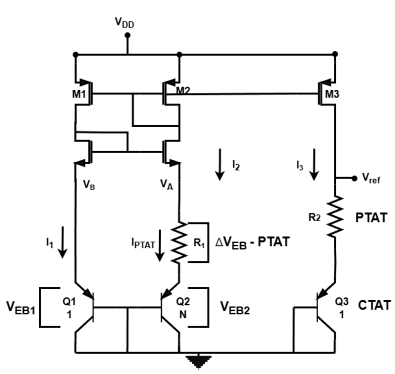

Figure 9. Current-Mirror Based BGR Topology

Transition to OTA-Based Design

Replacing the current mirror with an OTA (Fig. 8) improved performance significantly. Simulations with an ideal OTA showed that a gain of 60 dB yielded a PSRR of −43 dB. Gains beyond 80 dB provided marginal benefit and increased complexity.

Base-Emitter Area Ratio

A 1:8 base-emitter area ratio enabled a common-centroid layout that mitigates process gradients and temperature variations, balancing matching accuracy and layout area.

Resistor Selection for Temperature Stability

Rupoly resistors were selected for their low temperature coefficient (TC), critical for stable V_{ref}. Compared to diffusion or n-well resistors, Rupoly resistors minimize TC-induced current degradation at high temperatures.

Current Mirror Design and V_{ref} Considerations

Increasing transistor channel length L to $1\,\mu$m improved Early voltage V_A, raising output resistance r_{out} and stabilizing current against V_{DS} variations:

$$I_{DS,\text{out}} = I_{\text{ideal}}\left(1 + \frac{V_{DS}}{V_A}\right).$$

Matching drain voltages ($V_A \approx V_B \approx V_{\text{ref}}$) reduces channel-length modulation errors, boosting PSRR to −60 dB without complex cascoded mirrors. This also minimizes V_{ref} temperature coefficient with first-order compensation, producing $V_{\text{ref}} \approx 0.7\,\text{V}$, suitable for the target 0.9 V LDO output.

Operating Current Selection

An operating current of $\sim 5\,\mu$A was chosen to balance PSRR, power consumption, startup reliability, and noise. Lower current improves PSRR by increasing $r_{ds} = V_A/I_{DS}$, but very low currents increase susceptibility to leakage and degrade matching.

Resistor Sizing for Temperature Compensation

For a PTAT current $I_{\text{PTAT}} = 5\,\mu$A:

$$R_1 = \frac{V_T \ln(N)}{I_{\text{PTAT}}} \approx 10.98\,k\Omega,$$

with $N = 8$. $R_2 = 121.6\,k\Omega$ generates a complementary CTAT current; $R_3 = 66.63\,k\Omega$ scales $V_{\text{ref}} \approx 0.72\,\text{V}$:

$$R_3 = \frac{V_{\text{ref}}}{\frac{\ln(N)V_T}{R_1} + \frac{V_{BE}}{R_2}}.$$

Simulations confirmed this yields optimal temperature stability.

Final Device Sizing

By scaling V_{ref} down to $\sim 0.72\,\text{V}$, transistor lengths were reduced from $1\,\mu$m to 250 nm (with dummies) without compromising PSRR or line regulation, saving layout area.

Component	Type	Width (μm)	Length (μm)	Function
M0, M1, M2	PMOS (pch_18_mac)	1.8	0.25	Current mirror transistors
M3	PMOS (pch_18_mac)	0.9	0.25	Dummy transistor
M4	PMOS (pch_18_mac)	3.6	0.25	Dummy transistor
C1	PNP5 BJT	–	–	BGR diode (area: 2.5×10^{-11} m^2)
C2	PNP5 BJT	–	–	BGR diode (area: 2.0×10^{-10} m^2)
R3, R4	Rupoly resistor	0.43	90	CTAT resistors ($\sim 121.6\,k\Omega$)
R6	Rupoly resistor	0.4	46	Output scaling resistor ($\sim 66.63\,k\Omega$)
R7	Rupoly resistor	0.53	10	PTAT resistor ($\sim 10.98\,k\Omega$)

Table 2
Component Dimensions for BGR Circuit

3.3 OTA

Establishing Initial Design Specifications:

The design process refers to the OTA in Figure 3. The initial design of the OTA was performed early in the year, at a time when the precise circuit specifications were still unclear. To develop a baseline design with reasonable performance, we used a textbook methodology for LDO design as described in [15], using preliminary specifications as a starting point for future refinement. The initial design specifications included a DC gain of at least 60 dB, a gain-bandwidth product (GBW) in the range of 10–50 MHz, and a phase margin of approximately 60° to ensure loop stability. Since the OTA is used twice within the LDO, minimizing its power consumption was a key constraint, with a target of less than 200 μW.

Initially, the maximum load current was estimated at 8 mA, which dictated the sizing of the PMOS pass transistor. To estimate its parasitic gate capacitance, we performed a transient simulation (Figure 10) using a 0–1.8 V pulse source in series with a 500 kΩ resistor connected to the transistor's gate. The time constant τ was extracted at the point where the gate voltage reached 63.2% of its final value ($V = 0.632V_{\text{final}}$). From the relation:

$$\tau = RC \quad \Rightarrow \quad C = \frac{\tau}{R} = \frac{31.921\text{ ns}}{500\text{ k}\Omega} \approx 64\text{ fF},$$

an estimate for the gate capacitance of a PMOS with dimensions $W = 60\,\mu$m, $L = 150$ nm was obtained, which was the pass transistor size. This helped determine the required size for the compensation capacitor.

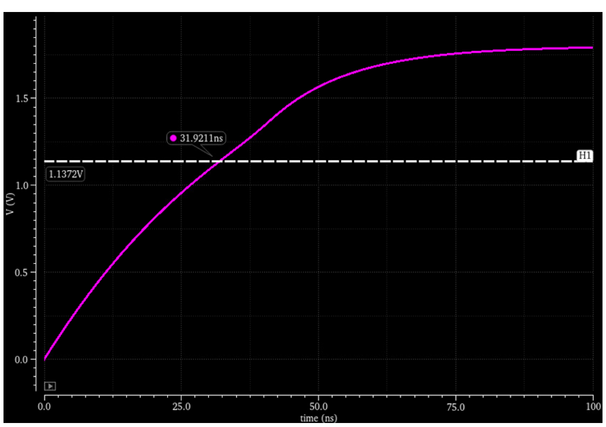

Figure 10. Finding the Parasitic Capacitance at the Gate of the Pass Transistor Through Simulation

To provide sufficient headroom, the load capacitor C_L was initially set to 100 fF. This value was sufficient not only for the OTA but also for the much smaller BGR current mirror transistors. However, mid-year, the maximum load current specification was increased to 55 mA, which required a significantly larger pass transistor. This increased the gate capacitance by roughly a factor of 10, to approximately 640 fF. Consequently, C_L was increased to 1 pF to maintain the desired frequency response. To preserve loop stability and maintain a good phase margin, the compensation capacitor was also increased, which in turn reduced the GBW by about half, but still within the acceptable design range.

The OTA's differential pair uses NMOS input transistors, and the expected common-mode input voltage was expected to be around $V_{\text{ref}} = 0.72$ V. While the input voltage is not expected to fall significantly below this level, it was important to ensure that all transistors remain in saturation, including in weak inversion. Therefore, the input common-mode range (ICMR−) was carefully minimized. The upper limit of the ICMR is not a concern, as it can extend up to 1.6 V without issue.

Slew rate (SR), which is the maximum rate of change of the OTA output, was not considered, since the OTA is used in low-speed analog blocks such as the BGR and LDO, where fast transients are uncommon.

Table 3 summarizes the initial OTA performance goals and the achieved results:

Specification	Initial Goal	Initial Results	Final Results - Pre Layout
DC Gain (dB)	>60	69.8	77.6
GBW (MHz)	5–30	23	6.4
ICMR (-)	Minimize	0.67 V	0.69 V
V_{out} Dynamic Range	Rail-to-Rail	0.1 V–1.7 V	0.1 V–1.7 V
Phase Margin (deg)	>60	72	76.5
C_{Load} (pF)	$0.1 \rightarrow 1$	0.1	1
C_C (pF)	>0.4	1	1.5
Power (μW)	<200	120	37

Table 3
OTA Design Specifications

OTA First Stage Design

The OTA consists of two stages. We begin by designing the first stage, which includes the biasing branch, tail transistor, differential input pair, and active load (current mirror) transistors. The second stage is initially modeled as a 1 pF capacitive load. The design targets for the first stage are to achieve approximately 40 dB gain, GBW of 30 MHz, and bias current of around 20 μA.

The transistor sizing for proper differential operation is:

$$\begin{aligned} M_1 &= M_2 \quad \text{(input NMOS)},\\ M_3 &= M_4 \quad \text{(PMOS current mirror)},\\ M_5 &= M_8 \quad \text{(bias and tail NMOS)}. \end{aligned}$$

Active Load Transistor Sizing (M_3, M_4): The sizes of the PMOS current mirror transistors M_3, M_4 are determined based on the positive input common-mode range (ICMR$^+$). To keep M_3, M_4 in strong inversion and saturation, we require:

$$V_{\text{DS}} > V_{\text{GS}} - V_{\text{TH}}.$$

Assuming the maximum input voltage is 1.2 V (arbitrary), the drain voltage of M_1 must be:

$$V_{D1} > V_{in} - V_{TH} = 1.2\,\text{V} - 0.5\,\text{V} = 0.7\,\text{V}.$$

To provide margin, we aim for $V_{D1} \approx 1\,\text{V}$.

Simulating a diode-connected PMOS with $W = 10\,\mu\text{m}, L = 1\,\mu\text{m}$, and bias current $I = 10\,\mu\text{A}$, we estimate:

$$\mu_p C_{ox} \approx 100\,\mu\text{A/V}^2, \quad V_{TH} \approx 0.5\,\text{V}.$$

For a current of $I_3 = 10\,\mu\text{A}$ and $V_{GS} \approx 0.8\,\text{V}$, the size ratio is:

$$\left(\frac{W}{L}\right)_{3,4} = \frac{2I_3}{\mu_p C_{ox}(V_{GS} - V_{th})^2} = \frac{20\,\mu}{100\,\mu \cdot (0.3)^2} \approx 2.2.$$

We therefore choose:

$$\left(\frac{W}{L}\right)_{3,4} = \frac{6\,\mu}{2\,\mu} = 3.$$

The relatively long channel length helps increase the output resistance and thus the stage gain.

Input Transistor Sizing (M_1, M_2): The sizing of the NMOS input pair is based on the GBW requirement. Assuming a single-pole first stage and DC gain of:

$$A_{v1} = g_{m1} \cdot (r_{o2} \parallel r_{o4}),$$

the pole is located at:

$$\omega_p = \frac{1}{C_L(r_{o2} \parallel r_{o4})}.$$

Thus, the gain-bandwidth product (GBW) becomes:

$$\text{GBW} = \frac{g_{m1}}{2\pi C_L}.$$

For a target GBW of 30 MHz and $C_L = 1\,\text{pF}$:

$$g_{m1} = 2\pi \cdot 1\,\text{pF} \cdot 30\,\text{MHz} = 188.5\,\mu\text{S}.$$

Extracting from simulation $\mu_n C_{ox} \approx 1\,\text{mA/V}^2$, and $I_{D1} = 10\,\mu\text{A}$, we solve:

$$\left(\frac{W}{L}\right)_{1,2} = \frac{g_{m1}^2}{2I_{D1}\mu_n C_{ox}} = \frac{(188.5\,\mu)^2}{2 \cdot 10\,\mu \cdot 1\,\text{m}} \approx 1.77.$$

We round this to:

$$\left(\frac{W}{L}\right)_{1,2} = \frac{2.8\,\mu}{1.6\,\mu} \approx 1.75.$$

This sizing ensures the OTA meets its bandwidth and stability targets with sufficient gain.

Tail Current Source and Biasing Branch Design (M_5, M_8): The sizing of transistors M_5 and M_8 is determined based on the negative input common-mode range (ICMR⁻) requirement. Since the input voltage is expected to be around 0.72 V, we must ensure that M_5 remains in saturation even at low input voltages. For proper operation of M_1, the condition is:

$$V_{\text{in,min}} > V_{\text{DSAT}_5} + V_{\text{GS1}}.$$

To estimate V_{GS1}, we use the quadratic current relation in saturation:

$$(V_{\text{GS1}} - V_{\text{th}})^2 = \frac{2I_1}{\mu_n C_{\text{ox}} \frac{W}{L}} = \frac{20\,\mu\text{A}}{1\,\text{mA/V}^2 \times 1.75} \approx 0.0114,$$

$$\Rightarrow V_{\text{GS1}} \approx V_{\text{th}} + \sqrt{0.0114} \approx 0.5\,\text{V} + 0.107 \approx 0.6\,\text{V}.$$

If we target $V_{\text{in,min}} = 0.7\,V$, then $V_{\text{DSAT}_5} \leq 0.093\,\text{V}$, which is challenging at the original tail current of $10\,\mu A$. Hence we reduce the tail current slightly to $7.5\,\mu A$ and increase the size of M_5 accordingly.

Using the saturation current equation:

$$\left(\frac{W}{L}\right)_{5,8} = \frac{2I_5}{\mu_n C_{\text{ox}}(V_{\text{GS}} - V_{\text{th}})^2},$$

to get an overdrive voltage 50 mV, we compute:

$$\left(\frac{W}{L}\right)_{5,8} = \frac{2 \cdot 15\,\mu\text{A}}{1\,\text{mA/V}^2 \cdot (50\,\text{mV})^2} = \frac{30\,\mu}{2.5\,\mu} = 12.$$

Using simulations, the final value chosen was:

$$\left(\frac{W}{L}\right)_{5,8} = \frac{8.4\,\mu\text{m}}{0.48\,\mu\text{m}} \approx 17.5.$$

Resistor R_1 Sizing for Bias Current: To generate the bias current $I_8 \approx 15\,\mu\text{A}$, we select R_1 such that the gate voltage of M_8 provides the correct V_{GS}. Assuming the drain of M_8 is approximately $V_{\text{DD}} = 1.3\,\text{V}$ and its source is connected through R_1 to ground, we estimate:

$$V_{GS8} = V_D - V_S = V_{DD} - I_8 R_1 \approx 0.55\,\text{V}.$$

Substituting into the saturation current equation:

$$I_8 = \frac{\mu_n C_{ox}}{2}\frac{W}{L}(1.25 - I_8 R_1)^2 = 8.75m(1.25 - 15\mu \times R_1)^2 = 15\mu,$$

hence, we get a quadratic equation on R_1, which gives us an initial estimation of $R_1 = 80\,k\Omega$. In simulation, a resistor of $70\,k\Omega$ gave us a current of around $15\,\mu A$ in the tail transistor.

After finalizing the transistor sizing, simulation confirmed that all devices remain in saturation across the expected input range. The input common-mode range (ICMR⁻) was verified to be around 0.67 V, while the upper bound exceeded 1 V, which is sufficient for the LDO. The first stage gain was measured to be approximately 36 dB, which is slightly below the design target of around 40 dB.

OTA Second Stage Design

We proceeded to the second stage, which is a common-source (CS) amplifier designed to boost the overall gain. To achieve a large output swing and improve common-mode rejection, this stage is implemented using a PMOS transistor. The gain of the second stage is given by:

$$A_2 = g_{m6}(r_{ds6} \parallel r_{ds7}).$$

The second stage introduces an additional pole located at:

$$p_2 = \frac{g_{m6}}{C_L},$$

where g_{m6} is the transconductance of the second-stage transistor, and C_L is the load capacitance, which at first was estimated to be 0.1 pF.

As a result of adding this stage, the circuit is no longer a single-pole system, and compensation is required to control the pole locations. Thus, we introduce a compensation capacitor, C_C. Its addition also introduces a right-half plane (RHP) zero at:

$$z_{\text{RHP}} = \frac{g_{m6}}{C_C}.$$

As discussed in the theory, there's Miller effect on the compensating capacitor, which shifts the first-stage pole to a lower frequency:

$$p_1 = \frac{1}{(r_{ds1} \parallel r_{ds3})A_2 C_C}.$$

This shift determines the gain-bandwidth product (GBW), which is now:

$$\text{GBW} = \text{DC gain} \times p_1 = \frac{g_{m1}}{C_C}.$$

Frequency Compensation and Phase Margin Optimization: Due to the added left-half plane (LHP) pole and the RHP zero, the phase margin is expected to decrease. To improve it, both the non-dominant pole and the RHP zero must be pushed to higher frequencies. The zero is placed at a frequency one decade beyond the GBW:

$$z = 10 \times \text{GBW}.$$

This implies:

$$\frac{g_{m6}}{C_C} = \frac{10 g_{m1}}{C_C} \quad \Rightarrow \quad g_{m6} = 10 g_{m1}.$$

To determine where to place the non-dominant pole, we must consider the desired phase margin. The phase of the transfer function at the gain-bandwidth (GBW) frequency, which approximately corresponds to the 0 dB frequency in a well compensated design, is given by:

$$\angle \frac{v_{\text{out}}}{v_{\text{in}}}(\text{GBW}) = -\tan^{-1}\left(\frac{\text{GBW}}{z}\right) - \tan^{-1}\left(\underbrace{\frac{\text{GBW}}{p_1}}_{A_{\text{DC}}}\right) - \tan^{-1}\left(\frac{\text{GBW}}{p_2}\right).$$

Assuming the zero is placed at $z = 10 \times \text{GBW}$ and A_{DC} is very large ($\tan^{-1}(A_{\text{DC}}) \approx 90°$), we simplify:

$$\text{Phase Margin} = 180° - \underbrace{\tan^{-1}(1/10)}_{\approx 5.71°} - 90° - \tan^{-1}\left(\frac{\text{GBW}}{p_2}\right).$$

To maintain a phase margin of at least 60°, solving for p_2 yields:

$$p_2 \geq 2.2 \times \text{GBW}.$$

Since $p_2 = \frac{g_{m6}}{C_L}$ and $p_1 = \frac{g_{m1}}{C_C}$, the inequality can be rewritten as:

$$\frac{g_{m6}}{C_L} \geq 2.2 \times \frac{g_{m1}}{C_C}.$$

Because $g_{m6} = 10 g_{m1}$ is costly in terms of power, an alternative approach was employed: a series resistor in the compensation network was added to shift the zero from the RHP to the left-half plane (LHP), providing beneficial positive phase shift and relaxing the conditions. This allows a smaller transconductance ratio of:

$$g_{m6} = 2\text{–}3 \times g_{m1}.$$

Given this, the phase margin condition becomes:

$$C_C \geq 0.74 \times C_L = 74\,\text{fF},$$

yielding a phase margin close to 60°.

To increase the phase margin to about 70°, the zero should be placed further above the GBW:

$$p_2 \geq 3.93 \times \text{GBW} \quad \Rightarrow \quad C_C \geq 1.3 \times C_L = 130\,\text{fF}.$$

Increasing C_C decreases the GBW, highlighting the classic trade-off between bandwidth and phase margin.

A compromise value of $C_C = 300\,\text{fF}$ was chosen, ensuring a robust phase margin of about 75°.

Sizing M_6 and M_7: The sizing of the gain transistor M_6 was determined from:

$$\frac{I_6}{I_4} = \frac{(W/L)_6}{(W/L)_4} = \frac{g_{m6}}{g_{m4}},$$

where

$$g_{m6} \approx 3 g_{m1} = 300\,\mu S, \quad g_{m4} = 67\,\mu S.$$

Given $(W/L)_4 = 3$, the aspect ratio is:

$$(W/L)_6 = \frac{300}{67} \times 3 \approx 13.4,$$

and the chosen practical sizing was:

$$\frac{6\,\mu m}{0.48\,\mu m} = 12.5.$$

The load transistor M_7 belongs to the current mirror with M_5 and M_8. Its sizing was chosen similarly:

$$(W/L)_7 = \frac{I_7}{I_5} \times (W/L)_5 = \frac{32\,\mu A}{15\,\mu A} \times \frac{8.4\,\mu m}{0.48\,\mu m} \approx 37.$$

Because the original channel length 0.48 μm yielded insufficient gain, M_7 channel length was increased to 13 μm, which reduced threshold voltage and allowed width reduction. The final sizing was:

$$(W/L)_7 = \frac{135\,\mu m}{13\,\mu m}.$$

The complete sizing table for the OTA used in the BGR circuit is shown in Table 4.

Component	Type	Width (μm)	Length (μm)	Multiplier	Purpose
R2	Resistor (rupolym_m)	0.4	73.92	–	Bias resistor (~ 107.06 $k\Omega$)
R3	Resistor (rupolym_m)	0.4	69	–	Bias resistor (~ 99.94 $k\Omega$)
R4	Resistor (rupolym_m)	0.4	8	–	Compensation resistor (~ 8.73 $k\Omega$)
M5	NMOS (nch_18_mac)	4	1.4	2	Tail transistor of diff pair
M8	NMOS (nch_18_mac)	4	1.4	2	Bias transistor in current mirror
M7	NMOS (nch_18_mac)	4	1.4	4	2nd stage active load transistor
M1, M2	NMOS (nch_18_mac)	6	1.8	1	Differential input pair
M3, M4	PMOS (pch_18_mac)	6	2.0	1	Active load of diff pair (current mirror)
M6	PMOS (pch_18_mac)	8	0.58	1	Second stage common-source amplifier
C1	Capacitor (cfmom_2t)	–	–	–	Compensation capacitor (1 pF)
C2	Capacitor (cfmom_2t)	–	–	–	Compensation capacitor (132.58 fF)
C5	Capacitor (cfmom_2t)	–	–	–	Compensation capacitor (372.14 fF)
M10	NMOS (nch_18_mac)	6	1.4	1	Dummy transistor (lower current mirror)
M14	NMOS (nch_18_mac)	1	1.4	8	Dummy transistor (lower current mirror)
M15	NMOS (nch_18_mac)	6	1.4	1	Dummy transistor (lower current mirror)
M17	NMOS (nch_18_mac)	1	1.4	8	Dummy transistor (lower current mirror)
M11	PMOS (pch_18_mac)	8	2.0	2	Dummy transistor (upper current mirror)
M12	PMOS (pch_18_mac)	1	2.0	4	Dummy transistor (upper current mirror)
M9	NMOS (nch_18_mac)	1	1.8	4	Dummy transistor (common-centroid for diff pair)
M0	NMOS (nch_18_mac)	14	1.8	2	Dummy transistor (common-centroid for diff pair)

Table 4
Component Sizing Summary for OTA in BGR

The *Multiplier* column denotes the number of parallel unit devices per component, enabling effective width scaling while preserving small unit sizes to improve layout matching. Splitting of compensation capacitors and the bias resistor is solely for layout optimization.

LDO OTA Layout Adaptation and Bias Sharing

The LDO OTA's performance—gain, bandwidth, and phase margin—is nearly identical to the BGR OTA, with the main difference being transistor sizing. Transistor aspect ratios are maintained, but absolute dimensions are scaled down for a more compact, rectangular layout better suited for DPLL integration. The LDO OTA shares its bias point with the BGR OTA, eliminating the need for a separate bias circuit and saving area. Table 5 lists the complete OTA sizing used in the BGR.

Component	Type	Width (μm)	Length (μm)	Multiplier	Purpose
M1, M2	NMOS (nch_18_mac)	2	1.8	2	Differential input pair
M5	NMOS (nch_18_mac)	2	1.4	2	Tail transistor
M3, M4	PMOS (pch_18_mac)	2	1.5	1	Active load (1st stage current mirror)
M6	PMOS (pch_18_mac)	4	0.56	1	Second stage common-source amplifier
M7	NMOS (nch_18_mac)	2	1.4	4	Active load (2nd stage)
R4	Resistor (rupolym_m)	0.4	7	–	Compensation resistor (~ 10.18 $k\Omega$)
C1	Capacitor (cfmom_2t)	–	–	–	Compensation capacitor (1.66 pF)
M10	NMOS (nch_18_mac)	4	1.4	1	Dummy transistor (lower current mirror)
M14	NMOS (nch_18_mac)	1	1.4	4	Dummy transistor (lower current mirror)
M15	NMOS (nch_18_mac)	4	1.4	1	Dummy transistor (lower current mirror)
M17	NMOS (nch_18_mac)	1	1.4	8	Dummy transistor (lower current mirror)
M12	PMOS (pch_18_mac)	1	1.5	4	Dummy transistor (upper current mirror)
M11	PMOS (pch_18_mac)	4	1.5	2	Dummy transistor (upper current mirror)
M9	NMOS (nch_18_mac)	1	1.8	4	Dummy transistor (common-centroid for diff pair)
M0	NMOS (nch_18_mac)	12	1.8	2	Dummy transistor (common-centroid for diff pair)

Table 5
Component Sizing Summary for OTA in LDO

3.4 *Low-Dropout Regulator*

Pass Transistor Sizing The pass transistor significantly influences load regulation, governed by its transconductance g_m (see Eq. **??**). To support a maximum load current of 55.5 mA, a PMOS device operating in deep saturation with minimum channel length $L = 150$ nm was selected to maximize current density and maintain compact layout.

The transistor width W must be sufficient to deliver the load current without requiring an excessive overdrive voltage ($V_{SG} - V_{th}$), which could exceed the supply voltage V_{DD}. While larger W reduces dropout voltage ($R_{\text{DS,ON}} \propto \frac{1}{W}$) and improves g_m, it also increases silicon area and gate capacitance, potentially degrading phase margin and loop stability.

Thus, W was chosen to balance these trade-offs, ensuring:

- Support for 55.5 mA load current.
- Adequate phase margin (45–60°) for stable operation.
- Reasonable silicon area.

Using the saturation current model,

$$W \geq \frac{2I_{\text{Load}}L}{\mu_p C_{ox}(V_{SG} - V_{th})^2},$$

with $\mu_p C_{ox} \approx 60$–$70\,\mu\text{A/V}^2$ extracted from measurements, the required width was estimated between 550 and 700 μm. A value of 600 μm was selected after simulation tuning as the optimal trade-off.

Feedback Network Resistor Sizing The output voltage is set by the resistor ratio:

$$\frac{V_{\text{out}}}{V_{\text{ref}}} = 1 + \frac{R_{fb1}}{R_{fb2}}.$$

While the ratio fixes the DC output voltage, absolute resistor values impact:

- **Power consumption:** Lower values increase quiescent current.
- **Stability:** Large resistors with parasitic capacitances form low-frequency poles that can degrade phase margin.
- **Noise:** Larger resistors generate more thermal noise, reducing regulation quality.

Balancing these, resistor values $R_1 \approx 40\,\text{k}\Omega$ and $R_2 \approx 10\,\text{k}\Omega$ were chosen to minimize power draw while maintaining stability.

Output Filter Network Stability Simulations sweeping the output capacitor and optional series resistor indicated stable LDO operation for output capacitances from 100 fF to 10 pF. Settling times varied between 1 μs and 100 μs, with optimal transient response observed at approximately 150–300 fF, matching the estimated load from decoupling capacitors in the DPLL blocks.

Enable Signal Implementation To extend functionality, two control signals were added:

- **EN**: Enables/disables the LDO. When `EN = 0`, output is forced to 0 V.
- **BYPASS_EN**: Bypasses the LDO output directly to supply (1.8 V) when `BYPASS_EN = 1`.

Two transistors act as switches at the output node:

- NMOS pulldown for $V_{\text{out}} = 0$ V when `EN = 0`.
- PMOS pullup for $V_{\text{out}} = 1.8$ V when `BYPASS_EN = 1`.

A third PMOS transistor shorts the pass transistor gate to its source to disable it during bypass or shutdown, preventing unwanted current draw.

Nominal regulation at 0.9 V occurs when EN = 1 and BYPASS_EN = 0. The mutually exclusive control states and resulting output voltages are summarized in Table 6.

EN	BYPASS_EN	Vout	$\text{GATE\_EN} = \text{EN} \cdot \overline{\text{BYPASS\_EN}}$	Description
0	0	0 V	0	Output pulled low by NMOS, pass transistor disabled.
0	1	Undefined	0	Contention between pullup and pulldown; avoid.
1	0	0.9 V	1	Normal operation; pass transistor enabled.
1	1	1.8 V	0	Output driven high by PMOS; pass transistor disabled.

Table 6
Enable Signal Logic and Output Behavior

The gate enable signal is implemented as

$$\text{GATE\_EN} = \text{EN} \cdot \overline{\text{BYPASS\_EN}},$$

with inverted control signals driving the pullup and pulldown transistors accordingly (Fig. 11). The logic gates and switching transistors occupy minimal layout area.

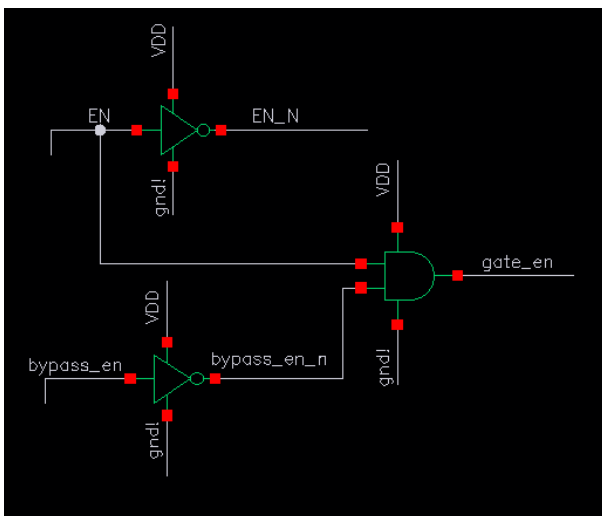

Figure 11. Enable Signal Logic Implementation

Summary of Device Sizing Table 7 lists key device dimensions in the LDO circuit.

Component	Type	Width (μm)	Length (μm)	Function
M0	PMOS (pch_18_mac)	600	0.15	Pass transistor
M3	PMOS (pch_18_mac)	10	0.15	Gate pull-up (disable pass)
M4	PMOS (pch_18_mac)	25	0.15	Output pull-up (bypass)
M2	NMOS (nch_18_mac)	0.3	0.5	Output pull-down (disable)
R1	Resistor (rupolym_m)	0.4	28.5	Feedback (~ 41.4 $k\Omega$)
R16	Resistor (rupolym_m)	0.53	9.5	Feedback (~ 10.4 $k\Omega$)
Output Cap.	Capacitor (digital block)	–	–	Output filter (100–300 fF)
Series Resistor	Resistor (digital block)	–	–	ESR (5–50 mΩ)

Table 7
Device Sizing Summary for LDO

3.5 Layout Implementation

Analog IC layout differs fundamentally from digital design, emphasizing precise matching, symmetry, parasitic control, and robustness against process and environmental variations. Unlike digital layouts focused on automated standard cell placement and timing, analog layouts require meticulous manual planning of device placement, symmetrical routing, and guard rings to ensure stable, predictable operation.

This section summarizes the layout flow for the Bandgap Reference, OTA, and LDO blocks, highlighting design constraints, matching strategies, and routing considerations.

OTA Layout The OTA employs a two-stage architecture with a differential input stage highly sensitive to mismatch and parasitics. Complete elimination of parasitics is impossible; however, ensuring matched devices experience identical parasitic environments minimizes their impact. Key strategies include:

- **Dummy Transistors:** Surround matched devices with dimensionally identical dummy transistors (all terminals shorted) to equalize parasitic capacitances and resistances.
- **Symmetric Routing:** Route critical interconnects and vias symmetrically to balance parasitic effects.

M3 and M4 serve as PMOS active loads ensuring current equality. Their layout (Fig. 12) exhibits mirror symmetry with dummy devices to balance parasitics.

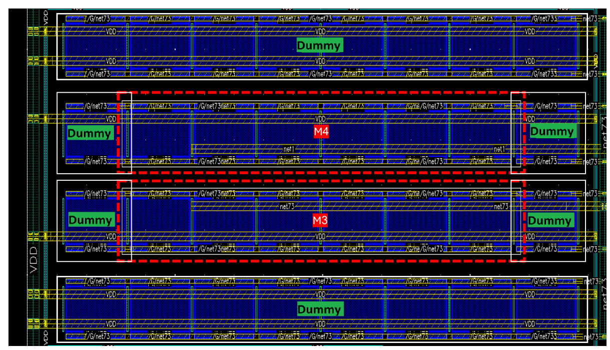

Figure 12. Layout of M3 and M4 with Dummy Devices and Symmetric Routing

M1 and M2 form the differential input pair, requiring very high matching. A common-centroid layout with split and interleaved devices and H-shaped routing (Fig. 13) enhances symmetry and reduces systematic mismatch, also mitigating wafer doping gradients.

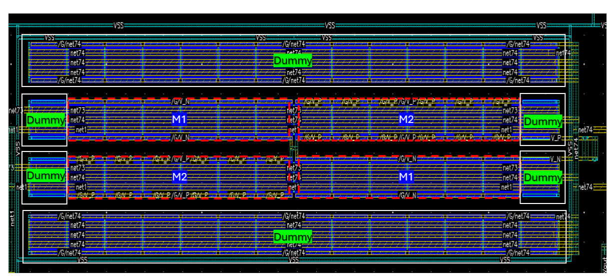

Figure 13. Common-Centroid Layout of Differential Pair with Symmetric H-shaped Routing

M5, M7, and M8 compose the current mirror biasing the OTA. Though less critical than the differential pair, symmetrical layout and careful routing preserve designed current ratios and amplifier performance.

The OTA floorplan (Fig. 14) aligns devices from V_{DD} to V_{SS} following natural current flow, optimizing routing and minimizing parasitic resistances and capacitances.

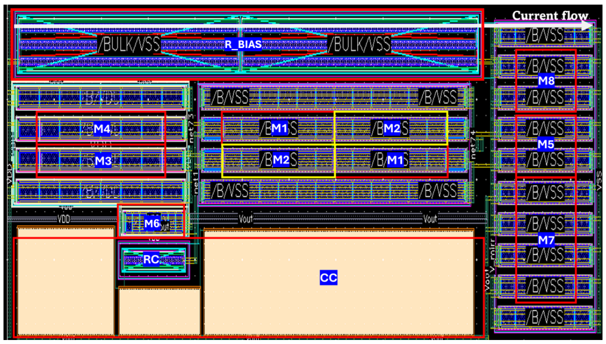

Figure 14. OTA Floorplan showing Functional Block Arrangement

Bandgap Reference Layout The Bandgap Reference uses diode-connected PNP transistors with a 1:8 size ratio to generate PTAT current. A common-centroid layout (Fig. 15) ensures matched process gradients, minimizing mismatch and enhancing temperature stability.

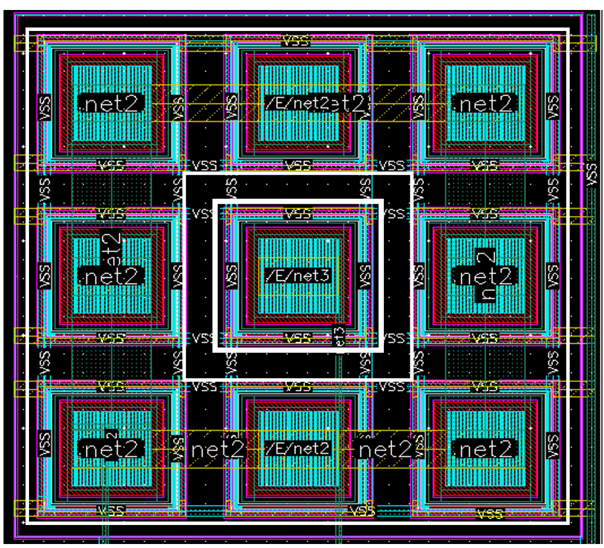

Figure 15. Common-Centroid Layout of Diode-Connected PNP Transistors

OTA Layout for LDO A separate OTA layout instance was created for the LDO to optimize silicon area and achieve a more square block, preserving device parameters and matching rules (Fig. 16). Both OTAs share a biasing network, saving area and current.

LDO Layout The PMOS pass transistor, sized 600 μm wide to support 55.5 mA load current, is segmented into 80 fingers across 5 instances to improve current distribution and reduce crowding.

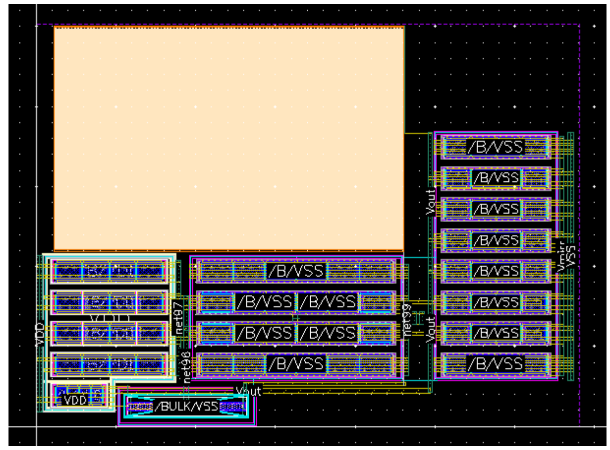

Figure 16. Modified OTA Layout for LDO: Compact and Square with Shared Biasing

Physical placement directs current flow from the V_{DD} input (lower left) to the regulated output (upper right), minimizing hot spots.

Routing employs a **metal stacking** approach, utilizing M2 to M8 layers to handle high current densities per PDK electromigration guidelines. Wide metal connections distribute current evenly (Fig. 17).

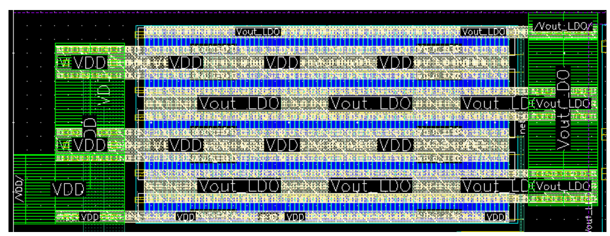

Figure 17. PMOS Pass Transistor Layout with Metal Stacking and Current Spreading

Final Integration The analog block occupies approximately 4699.7 μm^2, with metal fill density of 19.45% across M1–M9 layers, complying with DRC and supporting reliable current delivery. Power rails and sensitive references are aligned left-to-right (V_{DD} to V_{SS}) to minimize IR drops and thermal gradients. The integrated layout (Fig. 18) exhibits clean symmetry, ready for top-level integration.

4 Simulations

This section summarizes post-layout simulation results for the operational transconductance amplifier (OTA), bandgap reference (BGR), and low-dropout regulator (LDO). Schematic results are compared where relevant.

4.1 OTA

DC Operating Point The OTA employs a two-stage topology with the post-layout DC operating point shown in Fig. 19. The design features:

- **M1–M2:** Differential input pair.

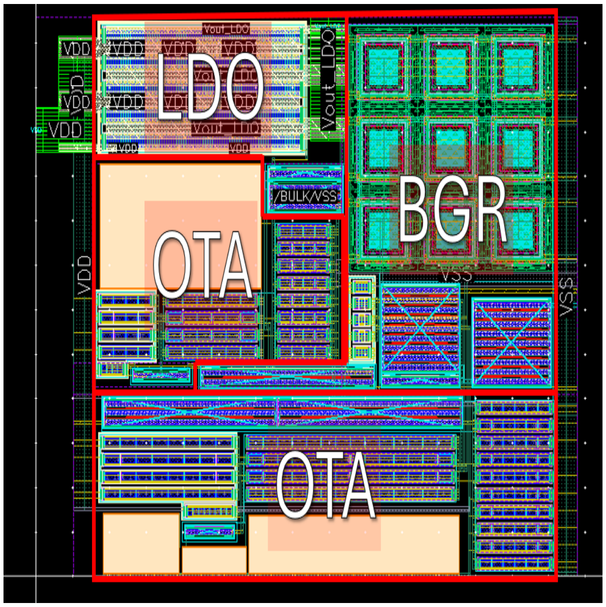

Figure 18. Final Integrated Layout of OTA, Bandgap Reference, and LDO

- **M3–M4:** Active load for first stage.
- **M6:** Common-source gain stage.
- **M5, M7, M8:** Bias current mirrors.
- **D1–D7:** Dummy devices for matching.
- Miller compensation network for stability.

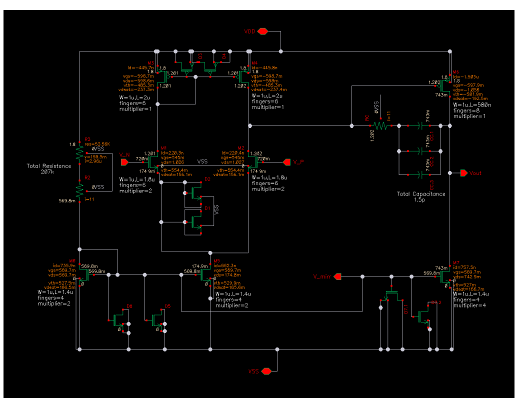

Figure 19. Post-layout DC operating point of the OTA.

AC Response The open-loop response (Fig. 20) shows:

$$\text{Gain}_{\text{DC}} \approx 77.5\ \text{dB}, \quad \text{UGF} \approx 5.6\ \text{MHz}, \quad \text{GBW} \approx 5.9\ \text{MHz}.$$

A dominant pole ensures stability with phase margin > 60°. High DC gain was prioritized to minimize steady-state offset in BGR and LDO loops, as their disturbance frequencies lie well below 100 Hz.

Output Swing A DC sweep from 710 mV to 730 mV on V_+ with $V_- = 720$ mV (Fig. 21) confirms near rail-to-rail output capability. In closed-loop operation, differential inputs remain in the mV range, but wide output swing is essential for driving the LDO pass device and ensuring regulation over all PVT conditions.

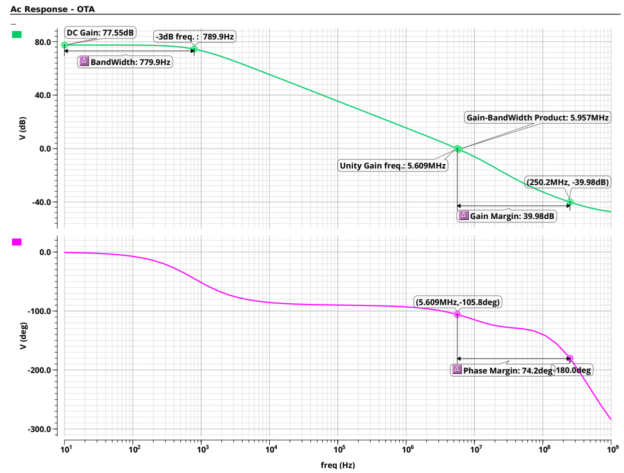

Figure 20. Open-loop AC response of the OTA.

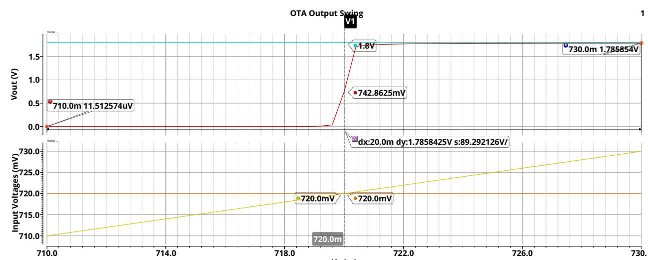

Figure 21. Output swing of the OTA, showing near rail-to-rail behavior.

4.2 Bandgap Reference (BGR)

The Bandgap Reference (BGR), shown in Fig. 22, consists of two diode-connected PNP BJTs (size ratio 1:8), an OTA to force equal branch voltages and generate a PTAT voltage, a PMOS current mirror, and resistors to weight PTAT and CTAT currents for temperature compensation. Unlike a conventional ≈ 1.2 V BGR, this modified topology targets $V_{\text{ref}} \approx 0.72$ V to meet the LDO's 0.9 V output requirement, while preserving design flexibility.

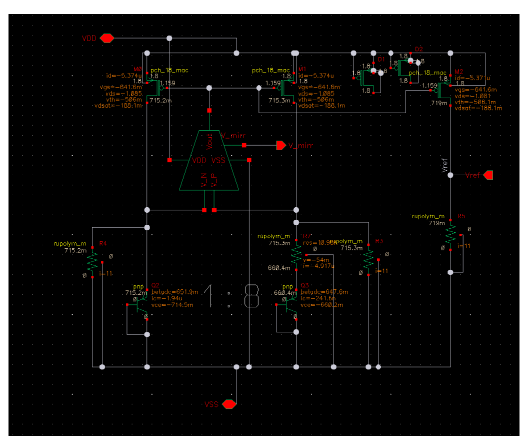

Figure 22. BGR schematic with annotated operating points.

Principle of Operation A PTAT voltage, V_{PTAT}, increases with temperature, while a CTAT voltage, V_{CTAT}, decreases. Weighted summation yields:

$$V_{ref} = \alpha V_{PTAT} + \beta V_{CTAT}$$

Choosing α and β for equal and opposite temperature coefficients minimizes V_{ref} variation (Fig. 23).

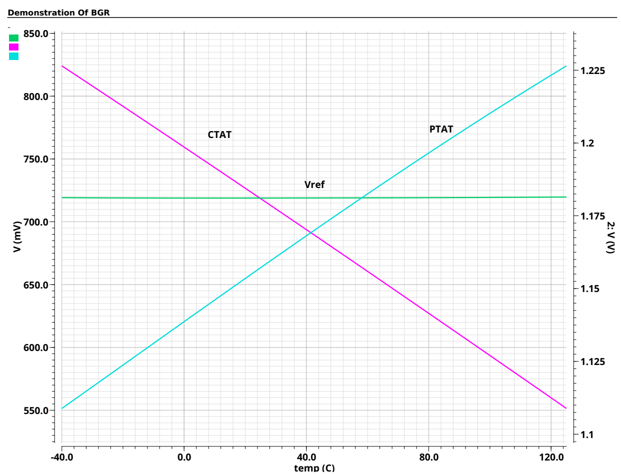

Figure 23. BGR concept: PTAT + CTAT → near-zero TC V_{ref}.

Performance Summary **Loop Gain:** Stable with large phase margin; low-frequency inversion from CS current mirror stage; high-frequency flattening due to pole-zero cancellation from the compensation network (Fig. 24).

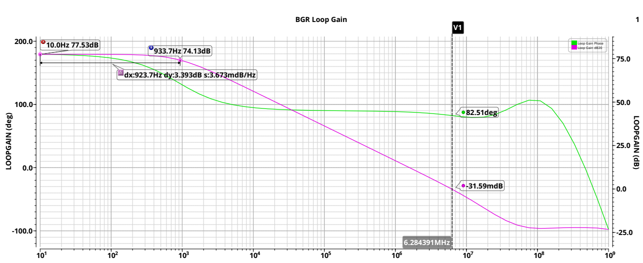

Figure 24. BGR loop gain simulation.

Temperature Coefficient: 4.6 ppm/°C pre-layout, 7 ppm/°C post-layout, with ≈ 2 mV drop due to parasitics (Fig. 25).

Line Regulation: Improved from 945.9 μV/V to 819.4 μV/V post-layout, possibly due to parasitic-induced bias shifts (Fig. **??**).

PSRR: Better than −60 dB up to 10 kHz, degrading to ≈ −20 dB at 1 MHz, acceptable with LDO output decoupling (Fig. **??**).

Startup: V_{ref} settles correctly across −40°C to 125°C for a 1 µs V_{DD} ramp (Fig. **??**).

This compact BGR achieves low TC, good supply independence, and robust startup, making it suitable as a precision reference for the integrated LDO.

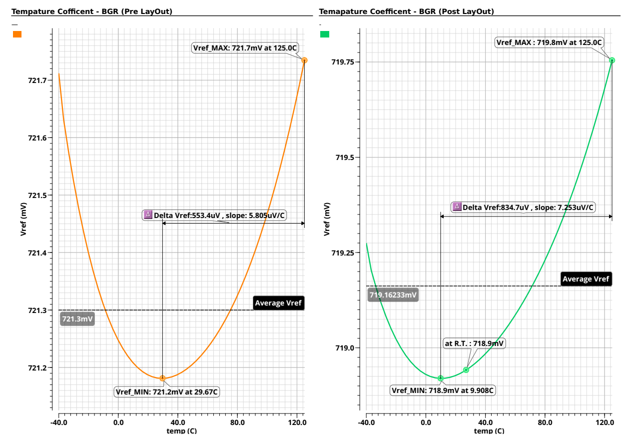

Figure 25. V_{ref} TC before/after layout.

4.3 LDO

The LDO consists of a Bandgap Reference (BGR), an Error Amplifier (OTA), a PMOS pass device, and a resistive feedback network, ensuring a stable output voltage across load and supply variations. The OTA compares the scaled output to the reference voltage, adjusting the PMOS gate to maintain regulation. The pass device consists of 400 fingers, with a total current capability of ≈ 60 mA.

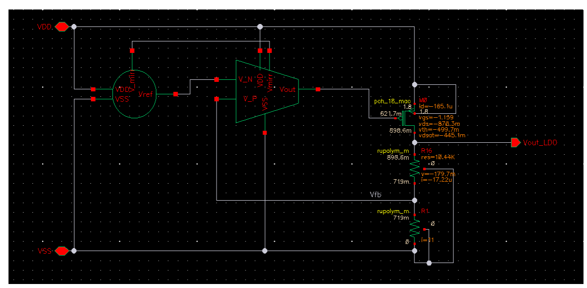

Figure 26. LDO schematic with annotated DC operating points.

Operating Principle A load current increase reduces V_{out}, which the OTA senses, driving the PMOS gate lower, increasing V_{SG} and restoring V_{out} (Fig. 27). Wide OTA output swing is critical for handling large load variations.

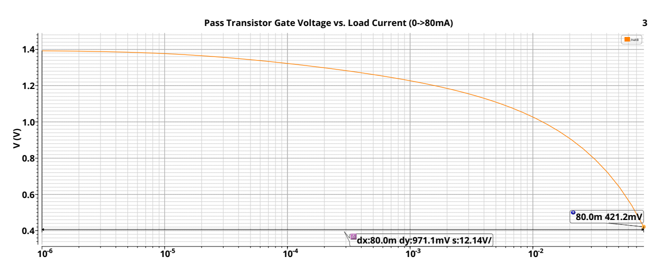

Figure 27. Gate modulation of pass transistor in response to load change.

Temperature Coefficient (TC) The post-layout TC increased from 5.1 to 6.4 ppm/°C (Fig. 28), closely tracking the BGR TC, as

V_{out} is locked to the reference voltage.

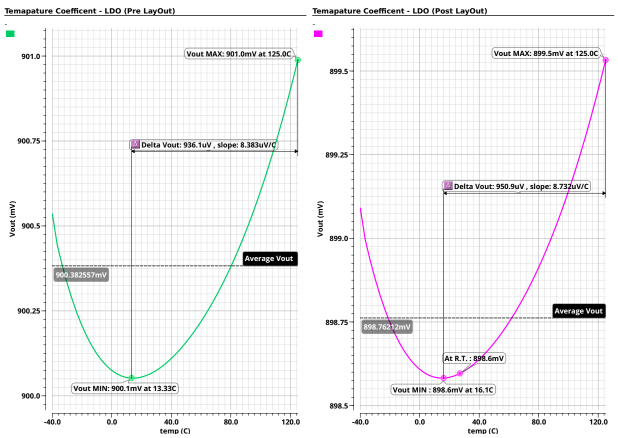

Figure 28. Pre- and post-layout LDO temperature coefficient.

Line and Load Regulation Line regulation degraded slightly from 774.4 to 893.8 μV/V post-layout, with $V_{DD,min}$ increasing from 1.08 to $\approx$ 1.3 V due to parasitics (Fig. **??**). Load regulation worsened from 2.25 mV/A to 32.7 mV/A, equivalent to 0.0327 Ω series resistance, mitigated via metal stacking.

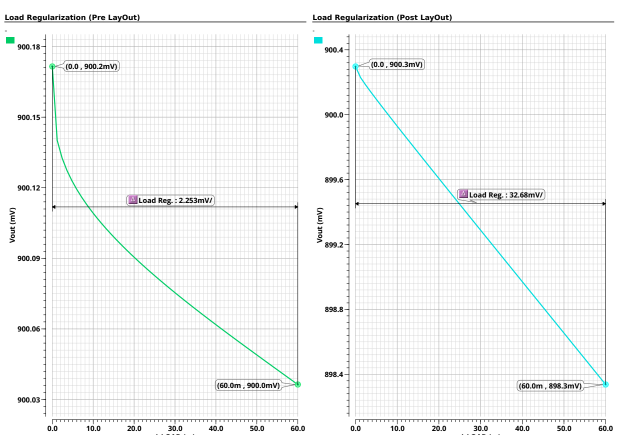

Figure 29. Pre- and post-layout load regulation.

Loop Gain and Stability Open-loop analysis (Fig. 30) shows a unity-gain bandwidth of 3.73 MHz and a phase margin of 57.8°, sufficient for stability across the operating range.

Transient Response Without output capacitance, the LDO cannot reject sub-ns VDD impulses (Fig. **??**), as its loop bandwidth is in the MHz range. With a 200 fF output capacitor, the instantaneous dip is reduced, improving DPLL supply resilience (Fig. 31).

Transient Response The LDO's post-layout transient performance was evaluated for ±55.5 mA load steps within 10 ps. Fig. 32 shows the step-down case (55.5 mA $\longrightarrow$ 0 mA $\longrightarrow$ 55.5 mA) with

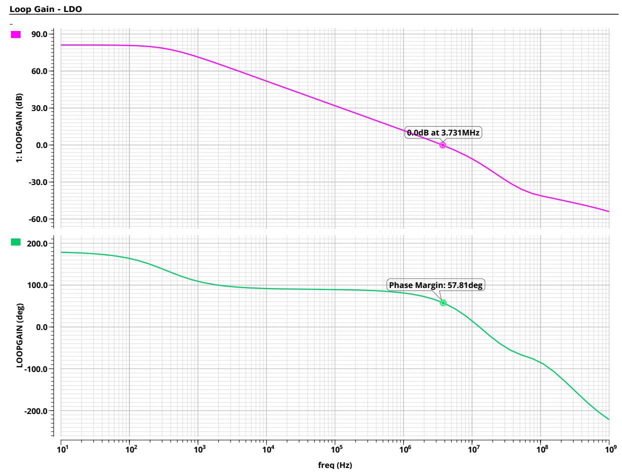

Figure 30. LDO loop gain and phase margin.

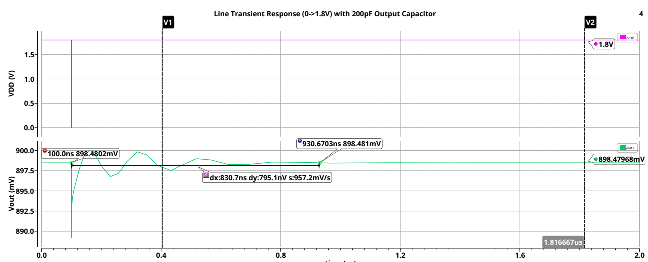

Figure 31. Transient response with 200 fF output capacitance.

minimal overshoot/undershoot and settling within 1 μs, indicating a well-damped loop. For the step-up case (0 mA $\longrightarrow$ 55.5 mA $\longrightarrow$ 0 mA), shown in Fig. 33, ringing is observed due to charge injection into C_{OUT} and parasitic L. Adding a small series resistor (5 mΩ) improved damping, but larger values ($\sim$ 5 Ω) caused unacceptable voltage dips ($\approx$ 0.4 V).

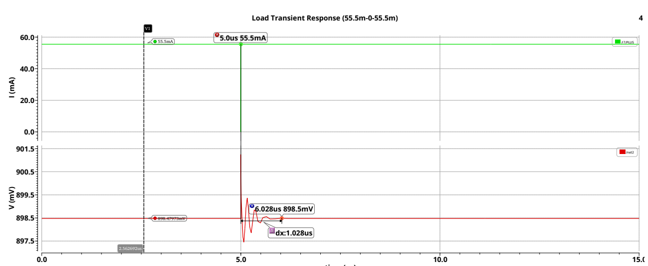

Figure 32. Post-layout transient response (55.5 mA $\longrightarrow$ 0 mA $\longrightarrow$ 55.5 mA).

PSRR The low-frequency PSRR remains below −60 dB (Fig. 34), limited at high frequency by OTA gain roll-off and LC resonance between C_{OUT} and parasitic L.

Output Noise The output noise density starts at $\approx$ 25 μV/$\sqrt{\text{Hz}}$ at low frequency and decreases to 672 nV/$\sqrt{\text{Hz}}$ at 100 kHz (Fig. 35), which is acceptable for DPLL loads.

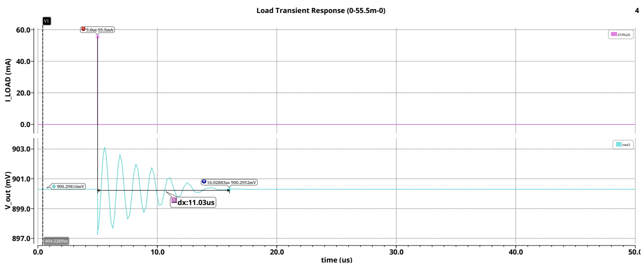

Figure 33. Post-layout transient response with 5 mΩ output resistor.

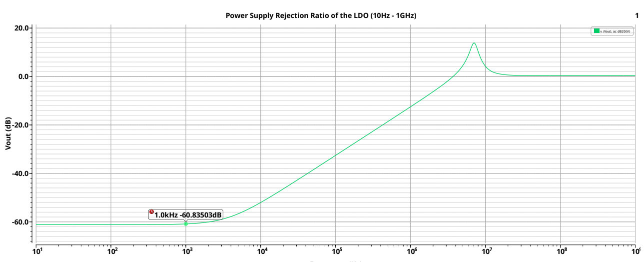

Figure 34. Post-layout PSRR (10 Hz–1 GHz), V_{DD} = 1.8 V, I_{Load} = 55.5 mA.

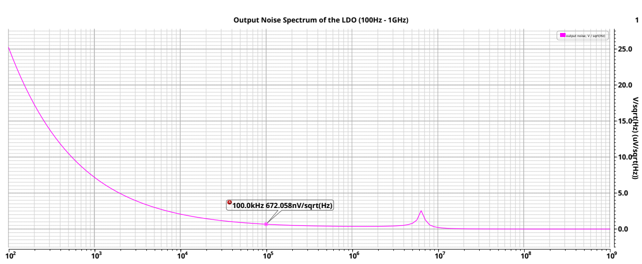

Figure 35. Post-layout output noise spectrum (100 Hz–1 GHz).

Quiescent Current and Efficiency The total current draw was $I_{\text{Total}} = 55.58$ mA with $I_Q \approx 80\,\mu$A, giving

$$\eta_{\text{current}} = \frac{55.5\text{ mA}}{55.58\text{ mA}} \approx 99.85\%.$$

With $V_{\text{OUT}} = 0.9$ V and $V_{DD} = 1.8$ V, the power efficiency is

$$\eta_{\text{power}} \approx \frac{0.9 \times 55.5}{1.8 \times 55.58} \approx 49.93\%,$$

as expected for the large dropout voltage.

Line and Load Regulation Across PVT Line and load regulation were simulated at TT, FF, SS, FS, and SF corners at 27°C, plus FF at 125°C and TT at -40°C. Figs. 36 and 37 show the LDO maintains stable regulation across all conditions.

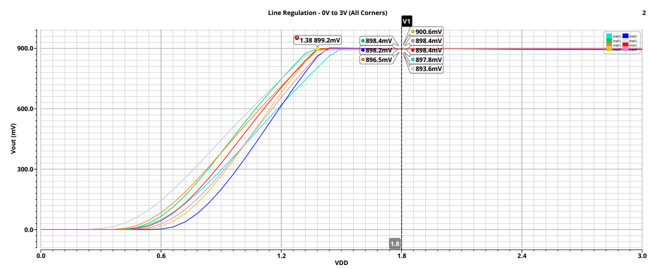

Figure 36. Line regulation across PVT corners.

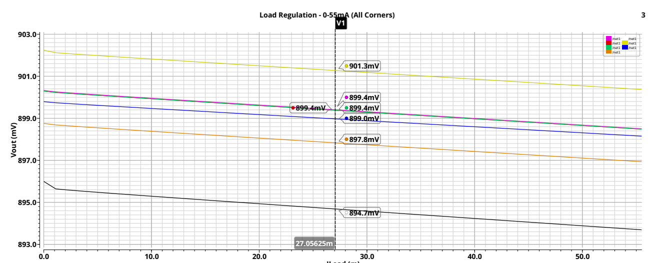

Figure 37. Load regulation across PVT corners.

LDO Operating Modes The LDO supports three modes, selected by `EN` and `bypass_en` control signals:

1) Disabled Mode (`EN = 0`, `bypass_en = 0`) The pass transistor and output pull-up are off, while an NMOS pulldown forces $V_{\text{OUT}} = 0$ V (Fig. 38). A gate pull-up shorts the pass transistor gate to V_{DD} to suppress leakage and static power.

2) Normal Regulation Mode (`EN = 1`, `bypass_en = 0`) The OTA feedback loop drives the pass transistor to regulate $V_{\text{OUT}} = 0.9$ V (Fig. 39). Output pull-up and pulldown devices are off; the gate pull-up is disconnected to allow OTA control.

3) Bypass Mode (`EN = 1`, `bypass_en = 1`) The regulated path is disabled, and V_{OUT} is directly connected to V_{DD} (1.8 V) via an active PMOS pull-up (Fig. 40). The pass transistor is held off by the gate pull-up; NMOS pulldown remains inactive.

Summary Table 8 summarizes device states in each mode, providing a compact view for implementation and verification.

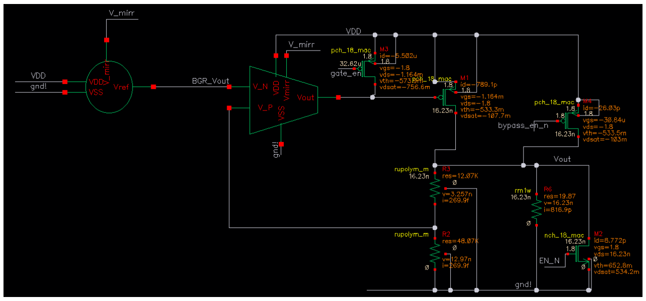

Figure 38. Disabled mode DC operating point.

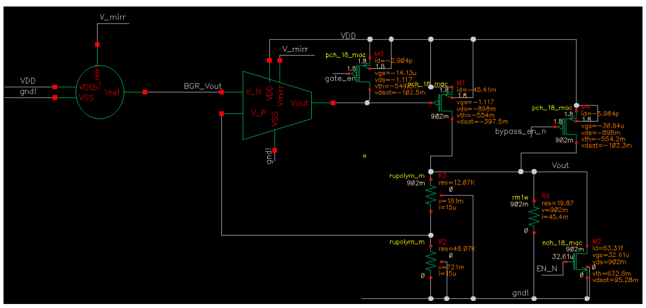

Figure 39. Normal operation DC operating point.

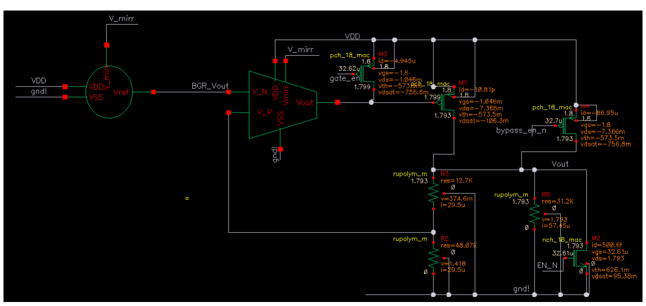

Figure 40. Bypass mode DC operating point.

Table 8
LDO operating modes and device states.

Mode	Pass FET	PMOS Pull-up	NMOS Pulldown
Disabled	OFF (gate=V_{DD})	OFF	ON
Normal	OTA-controlled	OFF	OFF
Bypass	OFF (gate=V_{DD})	ON	OFF

5 Results Analysis

This section summarizes the performance results of the designed LDO regulator and its sub-blocks, comparing pre-layout and post-layout simulations. Metrics include nominal output voltage, line regulation, load regulation, temperature coefficient, and PSRR, evaluated under the specified supply voltages and load conditions.

The Following tables compile these results to provide a concise overview of overall system behavior as well as the behavior of each sub-block,

Table 9 summarizes the OTA results pre- and post-layout.

Parameter	Spec	Pre-Layout	Post-Layout	Change (%)	Improvement over Spec
DC Gain (dB)	>70	77.6	77.5	−0.13 %	↑ 7.5 dB
BW (Hz)	N/A	816.7	789.9	−3.3 %	N/A
UGF (Hz)	≥5 M	6.27 M	5.6 M	−10.7 %	↑ 1.12×
GBW (Hz)	≥5 M	6.4 M	5.9 M	−7.8 %	↑ 1.18×
Phase Margin (deg)	>60	76.5	74.2	−3.0 %	↑ 14.2°

Table 9
OTA Performance Comparison Before and After Layout

Table 10 summarizes the BGR results pre- and post-layout.

Parameter	Spec	Pre-Layout	Post-Layout	Range	Change (%)	Improvement Over Spec
Temperature Coefficient (ppm/°C)	<60	4.6	7.03	−40–125 °C	+52.8 %	↑ 8.5×
Line Regulation (V/V)	<0.01	0.00058	0.00054	1.4–2.2 V	−6.9 %	↑ 18×

Table 10
BGR Performance Comparison Before and After Layout

Table 11 summarizes the LDO results pre- and post-layout.

Parameter	Spec	Pre-Layout	Post-Layout	Range	Change (%)	Improvement from Spec
Temperature Coefficient (ppm/°C)	< 60	5.1	6.4	−40–125 °C	+25.5 %	↑ 9.4×
Line Regulation (V/V)	< 0.01	0.0009	0.00089	1.4–2.2 V	−1.1 %	↑ 11.2×
Load Regulation (mV/mA)	< 0.1	0.0024	0.033	0–60 mA	+1275 %	↑ 3×
PSRR @ 1kHz	≤ −40 dB	-61 dB	-60.8 dB	N/A	−0.4 %	↑ 11×
Quiescent Current	≤ 1 mA	77 µA	80 µA	N/A	+3.75 %	↑ 12.5×

Table 11
LDO Performance Comparison Before and After Layout

Although performance variations can be observed after layout extraction in all measured parameters, these deviations remain within the acceptable design specifications. For example, the OTA bandwidth showed a modest decrease of approximately 3%, attributed to additional parasitic capacitances introduced by the layout, which is considered a minor effect in practical terms. Regarding the LDO line regulation, while values were simulated across a supply range of 1.4–2.2 V, the regulator has actually demonstrated correct operation at higher supply voltages as well, including up to 3 V, with stable behavior.

The most significant post-layout change was observed in the load regulation, which increased substantially. As explained in the implementation section, this is due to the inherent difficulty of maintaining precise regulation across high load currents, particularly given the unavoidable parasitic resistance of the pass transistor's metal routing, leading to IR drops no matter how well the layout is optimized.

Finally, although the LDO was developed primarily for a DPLL system, its performance and specifications would make it suitable for a variety of other analog or mixed-signal applications as well. Overall, the design remains compliant with its intended requirements and demonstrates robust operation across a wide range of conditions.

The following table compares the performance of the post-layout LDO with that of similar designs reported in the literature.

Parameter	This Work	[7]	[9]	[8]	[13]	[3]	[10]
Process	28nm	28nm	40nm	600nm	180nm	130nm	180nm
Layout Area (mm^2)	0.0044	0.0086	N/A	0.307	0.031	0.008	0.109
Supply Voltage V_{DD} (V)	1.38–3	0.9	1.1	1.5-4.5	1.2-1.8	1-1.4	1.31-3.3
Output Voltage V_{OUT} (V)	0.9	0.85	0.9	1.3	0.8-1.6	0.8	1.2
Load Current I_{Load} (mA)	55.5	20	100	100	100	25	50
Quiescent Current I_Q (μA)	80	33	30	N/A	10.2	112	5.5
Line Regulation (mV/V)	0.89	17.5	0.2	2.1	10	2.25	4.13
Load Regulation (mV/mA)	0.033	0.26	0.25	0.065	0.081	0.173	0.0056
Temperature Coefficient (ppm/°C)	6.4	N/A	N/A	38	N/A	N/A	87.5
PSRR (dB @1kHz)	-60.8	-24 @1MHz	-70 @10kHz	-60	N/A	-60	-36.3
Output Noise (nV/$\sqrt{Hz}$ @100kHz)	672	N/A	N/A	380	N/A	N/A	N/A
Settling Time - Worst Case (μs)	0.7	N/A	1.2	2	0.22	0.19	N/A
$C_L\,[pF]$	≈ 0.2	36	100	N/A	100	25	50
FOM [ps]	0.008	0.52	0.007	N/A	0.02	N/A	0.033

Table 12
Comparison of this LDO with LDO Implementations from the Literature

All values reported are post-layout, including full parasitic extraction, thereby ensuring the performance metrics reflect realistic behavior and confirming the suitability of the proposed design for practical DPLL integration.

Designed specifically for integration within a Digital Phase-Locked Loop (DPLL) system, this LDO is optimized for compactness, stability, and high load and line regulation. All essential for noise-sensitive mixed-signal environments. Additionally, contrary to other LDO implementations, this implementation can work with a very low output capacitances which come from the decoupling of the DPLL blocks.

The LDO achieves a layout area of only 0.0044 mm^2, making it one of the most compact among the compared works, which is particularly beneficial for dense SoC integration. Despite the small layout, it supports a load current of 55.5 mA, and meets the stringent load current specification of the DPLL.

It also exhibits strong regulation behavior, with a line regulation of 0.89 mV/V and load regulation of 0.033 mV/mA, while maintaining a temperature coefficient of 6.4 ppm/°C, which indicates excellent thermal stability. The quiescent current of 80μA is sufficiently low for low-power operation yet high enough to support fast transient response.

A key advantage is the PSRR of −60.8 dB at 1 kHz, which is especially important in DPLL applications where suppression of supply noise is critical for maintaining phase noise performance.

Compared to prior works, this LDO demonstrates highly competitive or superior results across all major metrics.

6 Conclusions and Further Work

Conclusions:

- The project successfully implemented a 0.9 V low-dropout (LDO) voltage regulator in 28 nm CMOS technology, which is designed for integration within a DPLL system.
- The LDO meets key performance specifications, including line regulation, load regulation, temperature stability, PSRR, and transient response, thereby validating both the circuit design and layout methodology.
- Layout-driven design considerations were critical in maintaining performance, with only minor degradations observed after post-layout extraction. This highlights the importance of early layout floorplanning and the application of industry-standard layout techniques.
- In general, the implemented LDO exhibits robust and reliable operation under the specified supply and load conditions across all PVT corners.

Further Work:

- **Error Amplifier Gain Improvement:** The EA gain can be improved beyond 80 dB using advanced topologies such as folded cascode with gain-boosting stages or telescopic cascode topologies, which can achieve gains of 100–120 dB. This can significantly improve line and load regulation, as well as power supply rejection ratio (PSRR). However, the telescopic cascode has limited output swing, which makes it more suitable for use as the error amplifier in the bandgap reference (BGR).
- **Error Amplifier Bandwidth Improvement:** The OTA bandwidth can be increased while maintaining a similar gain by adjusting the two-stage OTA parameters according to the desired performance. Specifically, the gain-bandwidth product (GBW), and consequently the bandwidth (BW), can be increased by boosting the transconductance (g_m) of the first stage. This can be achieved by increasing the OTA's bias current. In strong inversion, a tenfold increase in bias current roughly results in a tenfold increase in GBW, leading to a significant bandwidth improvement. However, to maintain proper pole splitting and stability, the second stage current should also be increased proportionally. Thus, this bandwidth enhancement comes at the cost of higher power consumption. This trade-off might be worth it at times since increasing the error amplifer's GBW improves power supply rejection ratio (PSRR) at higher frequencies by extending the frequency range over which the feedback loop effectively suppresses supply noise. However, in a 2-stage OTA increasing the bias current also reduces the output resistance (r_{ds}), which can eventually degrade the gain. If pushed too far, at some point, the gain loss due to lower r_{ds} outweighs the benefit from higher g_m. Therefore, a topology change to a cascode stage can effectively address this gain-bandwidth trade-off by stacking transistors to significantly increase output resistance by approximately a factor of $g_m r_{ds}$, which allows for high gain to be maintained even at elevated bias currents. Therefore, cascode stages offer a more favorable trade-off for LDOs where both gain and speed are critical.

- **BGR Improvements:** Second-order curvature-correction techniques can be implemented in the BGR to further reduce the temperature coefficient (TC), especially at higher temperatures. The TC and PSRR of the reference voltage can also be improved by using a cascode current mirror, which increases output impedance and can help stabilize the reference voltage against supply and process variations.
- **Pass Transistor Sizing and Buffering:** Load regulation can be improved by increasing the width of the pass transistor, which improves current drive capability and increases g_m. However, this increases the gate capacitance and shifts the pole there to lower frequencies, which can potentially cause stability issues. To mitigate this, a buffer stage can be added between the OTA output and the gate of the pass transistor. This buffer isolates the OTA from the capacitive load, and thus can improve bandwidth and contribute additional gain and improve PSRR. The trade-offs include increased power consumption, area, and added design complexity because of an additional pole.
- **LDO Function Extension:** The current design includes an enable signal to activate the LDO and produce 0.9V and another to bypass the LDO, to deliver 1.8 V directly to the DPLL blocks. Additional control signals can be introduced to enhance LDO functionality. These can be a tunable resistor ladder network in the feedback path for output voltage trimming using control bits, a high-impedance output mode, or a low-power operational mode for energy efficiency.
- **Noise Reduction:** Output noise can be reduced by utilizing lower-noise OTA topologies such as folded cascode instead of a two-stage OTA. Additionally, the intentional addition of an output capacitor in series with a small resistor can help filter high-frequency noise and suppress voltage spikes, at the price of degrading the transient response speed. Low-frequency noise, which is dominated by flicker noise, can be reduced by adding a passive RC filter network at the output of the Bandgap reference circuit, However, to filter out very low frequencies, large capacitors of size 1-10nF and resistors of size $1M\Omega$ are required, which are impractical on a chip.
- **Layout and Routing for Load Regulation:** Load regulation can be further enhanced through improved current routing, advanced current-splitting strategies, or better metal stack utilization. These layout optimizations might help reduce IR drop and improve the LDO's ability to handle high load currents efficiently.

7 Project Documentation

The complete project repository, including design files, deliverables, and figures, is available on GitHub at the following link:

`https://github.com/ArsenArutyunov/DPLL-LDO`

The repository is organized into the following main directories:

- **Weekly Presentation/**
  This folder documents the project's weekly progress. It is divided into two subfolders:
  - **Semester A/** – Contains presentation materials from the first semester.
  - **Semester B/** – Contains presentation materials from the second semester.
- **Deliverables/** This folder includes all formal submission items:
  - The final project book.
  - The final presentation.
  - The project poster.
  - The mid-project presentation.
  - The initial workplan.
- **Figures/** This folder contains visual materials, such as:
  - Circuit diagrams.
  - Simulation graphs.
  - Layout screenshots.
  - Devices Sizes Tables.
  - Results Tables.
- **Design/** This directory contains the full Cadence design environment for the Low Dropout Regulator (LDO), including schematics, layouts, testbenches, and simulation setups.

Acknowledgements

Special thanks to Pini Schnaider (Intel) for his additional guidance and valuable insights during the layout implementation phase.

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